



The implications of financial frictions and imperfect knowledge in the estimated DSGE model of the U.S. economy



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ABSTRACT

In this paper, I study how alternative assumptions about expectation formation can modify the implications of financial frictions for the real economy. I incorporate a financial accelerator mechanism into a version of the Smets and Wouters (2007) DSGE framework and explore the properties of the model assuming, on the one hand, complete rationality of expectations and, alternatively, several learning algorithms that differ in terms of the information set used by agents to produce the forecasts. I show that the implications of the financial accelerator for the business cycle may vary depending on the approach to modeling the expectations. The results suggest that the learning scheme based on small forecasting functions is able to amplify the effects of financial frictions relative to the model with Rational Expectations. Specifically, I show that the dynamics of real variables under learning is driven to a significant extent by the time variation of agents' beliefs about financial sector variables. During periods when agents perceive asset prices as being relatively more persistent, financial shocks lead to more pronounced macroeconomic outcomes. The amplification effect rises as financial frictions become more severe. At the same time, a learning specification in which agents use more information to generate predictions produces very different asset price and investment dynamics. In such a framework, learning cannot significantly alter the real effects of financial frictions implied by the Rational Expectations model.

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1. Introduction

In this paper, I contribute to studying the macro-financial linkages by focusing on adaptively formed expectations (adaptive learning) as a mechanism that can potentially amplify and propagate shocks to the real economy. The latest financial turmoil has demonstrated that the impact of imbalances in the financial sector on the real economy and wealth can be far more influential than many economists have anticipated. The failure of the DSGE models that were in use prior to the crisis, to capture and explain the link between financial and real markets has pointed to the problem of the models misspecification in this important dimension. Even the more recent studies that analyze DSGE models with financial frictions have documented the problems in replicating the observed dynamics, explaining the “surprising” origin of the crisis and its propagation channels.¹ Therefore, there is a need for further research in the field of macro-financial modeling. In particular, it might be helpful to consider additional features of the transmission mechanism that could potentially augment the

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¹ Brzoza-Brzezina and Kolasa (2013) present an empirical evaluation of the two most popular approaches to implementing financial frictions into DSGE models: the Bernanke et al. (1999) setup, and the Kiyotaki and Moore (1997) model. They find that models with the Bernanke et al. framework fit data

interactions between the financial and real sectors thus making models more successful in describing the business cycle dynamics. In this work, I study the interaction between the “expectations” and “macro-financial” factors exploring the fact that public’s predictions play an important role in driving financial markets. I assess how departure from the complete rationality assumption in modelling the expectations about forward-looking (in particular, financial) variables may impact agents’ decisions over time, in addition to changes already induced by the presence of financial frictions, and whether the time variation of agents’ predictions (beliefs) can improve the fit of the estimated model and its dynamic properties.

More specifically, I incorporate adaptively formed expectations (Evans and Honkapohja, 2001; Milani, 2007; Orphanides and Williams, 2007) into a version of the Smets and Wouters (2007) DSGE model with a financial accelerator (FA). It is assumed that agents know the structure but they are uncertain about the parameters of the model. To learn the parameters, they formulate linear econometric models based on their economic perceptions and re-estimate these models as soon as new information arrives. Following Slobodyan and Wouters (2012b), I assume that forecasting models are re-estimated every period on the basis of the Kalman filter learning algorithm. I estimate the model using Bayesian methods and assess the joint role of financial frictions and the departure from the complete rationality assumption for the U.S. business cycle. On the basis of the estimation and simulation results, I assess how adaptive learning (AL) algorithms can modify the transmission mechanism of the model with RE and investigate the ability of AL to generate additional macroeconomic fluctuations in line with real data.

To the best of my knowledge, this paper is the first one that evaluates the effects of financial frictions under AL within the estimated DSGE model. Caputo et al. (2010) study how a FA mechanism combined with AL influences the business cycle fluctuations in a calibrated model. Another important difference of their paper is the information set that learning agents are assumed to use in order to form their predictions. The results of Caputo et al. (2010) are derived from so-called “minimum-state-variable” (MSV) learning. This means that agents use the full set of endogenous and exogenous variables in their forecasting functions.² Authors demonstrate that the size of the (sequential) shocks and the degree of the financial fragility are crucial for the amplification of the responses to shocks under this form of learning. In this paper, I follow Slobodyan and Wouters (2012b) and assume that agents’ forecasts can be also based on very small forecasting models, in particular on a model where expected value of a forward-looking variable depends on a constant and two lags of this variable. In fact, I compare the results for alternative information sets and demonstrate that the learning scheme is an important determinant of the effects of the FA for the real economy.

The main result of my work suggests that learning based on small forecasting models can amplify the effect of financial frictions and strengthen the propagation of shocks to the real economy. In particular, I demonstrate that in a model with a FA and learning may create additional business cycle fluctuations by influencing the degree of economic persistence in asset prices and investments. Specifically, time-varying agents’ perceptions about the economic developments may translate into greater perceived asset price persistence and more pronounced (and persistent) responses of investment and output. Simulation results also demonstrate that the sensitivity of the economy to financial shocks increased after the middle of 1980s, thus strengthening the link between asset prices and the real economy.

Comparing the forecasting performance of the RE and the best-performing AL model, I find that the AL model is more successful in predicting inflation in the mid 1970s and the beginning of the 1980s (in line with Slobodyan and Wouters, 2012a and Milani, 2014). Output and investment growth is better predicted by the AL model during the recessions at the beginning of the 1980s and 1990s. In addition, the learning model is particularly successful in capturing the most recent economic downturn, characterized by a significant and persistent drop in the real sector.

Regarding the estimated parameters, I find that some of the estimated structural rigidities and shock persistence decrease. More specifically, autoregressive components of exogenous processes for price and wage mark-up shocks fall significantly, similar to Slobodyan and Wouters (2012a) and Milani (2014). The decline in the persistence of the shock to investment technology is not so dramatic, but still notable. In addition, the variance of the investment shock declines under learning. These results imply that learning represents an alternative source of endogenous inertia and is helpful in explaining inflation, wage and investment dynamics. The estimated financial frictions parameters tend to increase somewhat under learning, affecting the persistence of the external finance premium and strengthening the impact of the FA mechanism compared to the model with RE.

The results of my work are generally in line with the AL literature. A number of studies have already demonstrated that AL can improve the fit of macroeconomic models. In particular, Milani (2007, 2008) and Sargent et al. (2006) have shown that introducing AL can generate the levels of persistence observed in U.S. data. Slobodyan and Wouters (2012a) assess whether learning may overcome the problem of DSGE models misspecification, pointed up in Del Negro et al. (2007). They demonstrate that the best-performing learning model may closely approximate the optimal DSGE-VAR (in terms of marginal likelihood and impulse response functions), which proves that the model-consistent expectations imposed by the RE hypothesis are too restrictive. The authors also illustrate that specification of initial beliefs plays an important role for this result. Slobodyan and Wouters (2012b) incorporate less-than-rational beliefs into the Smets and Wouters (2007) model and find that the impact of AL on macro dynamics is more pronounced when the agents’ information set is more restrictive than

(footnote continued)

better. However, they point out that even these models do not make a clear improvement over the New Keynesian benchmark in terms of marginal likelihood and matching the VAR impulse responses.

² The same set of variables is used to form forecasts under RE.

the one implied by RE. In small forecasting models, learning can explain episodes of inflation dynamics in the U.S. and also lowers the persistence of some of the exogenous shocks. Rychalovska and Slobodyan (2010) estimate a set of small to medium-sized DSGE models for the Euro Area. They also find that assuming adaptive expectations results in better model fit than if RE is used, especially when the agents use very little information to form their beliefs. Therefore, the conclusion that AL based on small forecasting models outperforms MSV learning and RE models seems to be a robust one, at least for the U.S. and European data.

The rest of the paper is organized as follows: in Section 2, I present the model. In particular, I focus on modeling the financial sector and on explaining the expectation formation mechanism; Section 3 contains the estimation methodology and results; Section 4 describes the effects of financial frictions on the transmission mechanism in the model with AL, and Section 5 concludes.

2. The model with a financial accelerator under learning

In this paper, I incorporate financial frictions combined with an imperfectly rational expectation formation mechanism into a medium-scale DSGE model based on Smets and Wouters (2007). In the specification of the Kalman filter learning, I closely follow Slobodyan and Wouters (2012b). The financial frictions are integrated in the form of the FA as originally applied in Bernanke and Gertler (1989); Bernanke et al. (1999) and extensively explored in the recent literature. The most relevant examples include Christiano et al. (2010), De Graeve (2008) and Christensen and Dib (2008). The model contains a number of nominal and real rigidities such as monopolistic competition on goods and labor markets, Calvo price and wage stickiness, habit formation in consumption and capital adjustment costs. Following the seminal contributions of Smets and Wouters (2003, 2007) and Christiano et al. (2005), these structural rigidities have become widely used in order to match the observed properties of the main macroeconomic series.

The financial sector is represented by capital good producers, financial intermediaries and entrepreneurs. As in Bernanke et al. (1999), financial market imperfections are set up in the form of asymmetric information between entrepreneurs and the banks. Due to this friction, the optimal financial contract, which maximizes the entrepreneur's payoff subject to the required rate of return of lenders, implies the existence of an endogenous external finance premium that depends on the entrepreneur's leverage ratio. The financial frictions of Bernanke et al. (1999) are based on the idea that asset price variability affects the entrepreneurial financial position and therefore drives credit market imperfections. Introducing AL and incomplete information into such a framework brings in additional dynamics in asset prices and the external finance premium. As a result, imperfectly rational beliefs can become an important driving force behind the fluctuations on financial markets and modify the impact of financial frictions on the real economy.

The model is detrended with a deterministic trend γ that represents a labor-augmenting growth rate in the economy. The non-linear system is then linearized around the stationary steady state of the detrended variables. Lower-case variables denote detrended variables expressed in real terms. In this section, I focus mainly on describing the financial sector as well as the learning algorithm. For the formal presentation of the rest of the micro-foundations, I would like to refer to the original Smets and Wouters (2007) model.

2.1. Financial sector

2.1.1. Capital-goods producers

Following Bernanke et al. (1999) and Christiano et al. (2003), competitive capital-goods producers, owned by households, produce new capital goods which are sold to entrepreneurs at price Q_t . Capital-goods producers combine investment goods, purchased from the final good producers, with the existing capital stock, rented from the entrepreneurs, to produce new capital goods, K_{t+1} . As in Bernanke et al. (1999), it is assumed that the rental rate for the existing capital is zero, since the operation takes place within one period. Capital-goods producers are subject to quadratic adjustment costs specified as function $S\left(\frac{I_t}{I_{t-1}}\right)$, with $S''(\cdot) > 0$. In addition, the capital production technology is affected by an investment-specific shock ε_t^i . The optimization problem of capital-goods producers, in real terms, consists of choosing the level of investment I_t to maximize the real expected profits:

$$\max_{I_t} E_t \left\{ \sum_{s=0}^{\infty} \beta^s \frac{\lambda_{t+s}}{\lambda_t} \left[Q_{t+s} I_{t+s} \varepsilon_{t+s}^i - I_{t+s} - Q_{t+s} I_{t+s} \varepsilon_{t+s}^i S\left(\frac{I_{t+s}}{I_{t+s-1}}\right) \right] \right\}, \quad (1)$$

where λ_t denotes the marginal utility of the real income of the household. The solution to the problem is:

$$\varepsilon_t^i Q_t \left(1 - S\left(\frac{I_t}{I_{t-1}}\right) \right) = 1 + \varepsilon_t^i Q_t S'\left(\frac{I_t}{I_{t-1}}\right) \frac{I_t}{I_{t-1}} - E_t \left\{ \beta \frac{\lambda_{t+1}}{\lambda_t} \varepsilon_{t+1}^i Q_{t+1} S'\left(\frac{I_{t+1}}{I_t}\right) \left(\frac{I_{t+1}}{I_t}\right)^2 \right\}. \quad (2)$$

Eq. (2) relates the price of capital to investment and adjustment costs. In the absence of adjustment costs, Q_t is constant and

equal to one. After detrending and log-linearization of (2), the dynamics of investment are given by:

$$\hat{i}_t = \frac{1}{(1+\bar{\beta}\gamma)} \left(\hat{i}_{t-1} + (\bar{\beta}\gamma) E_t \hat{i}_{t+1} + \frac{1}{\gamma^2 S''} \hat{Q}_t \right) + \hat{q}_t, \quad (3)$$

where S'' is the steady-state elasticity of the capital adjustment cost function and $\bar{\beta} = (\beta/\gamma^{\sigma_c})$, where β is the discount factor applied by households. As in [Christiano et al. \(2005\)](#), a higher elasticity of the cost of adjusting capital reduces the sensitivity of investment ($\hat{i}_t$) to the real value of the existing capital stock ($\hat{Q}_t$). Finally, $\hat{q}_t$ represents a disturbance to the investment-specific technology process and is assumed to follow a first-order autoregressive process with an iid-normal error term: $\hat{q}_t = \rho_q \hat{q}_{t-1} + \epsilon_t^q$.

The evolution of the capital stock is represented by the following expression:

$$\hat{k}_t = \left(1 - \frac{i_k}{k_*} \right) \hat{k}_{t-1} + \frac{i_k}{k_*} \hat{i}_t + \frac{i_k}{k_*} (1 + \bar{\beta}\gamma) \gamma^2 S'' \hat{q}_t. \quad (4)$$

2.1.2. Entrepreneurs and banks

In the original [Smets and Wouters \(2007\)](#) model, financial markets do not incorporate endogenous forms of inefficiency. In particular, households can borrow in any quantity at the rate that might exceed the risk-free rate R_t set by the central bank due to the exogenous premium ε_t^b . Modeling endogenous credit imperfections requires distinguishing between borrowers and lenders and the existence of a conflict between the two parts. Therefore, new types of agents have to be introduced. Following [Bernanke et al. \(1999\)](#), entrepreneurs, who are risk neutral and survive until the next period with probability χ , use their own funds (the net worth, N_{t+1}) and loans from the bank (B_{t+1}) to finance capital that is rented to the production sector. Competitive banks finance the loans by accepting deposits from the households at the risk-free rate. The financial intermediation between the banks and entrepreneurs is subject to friction based on the agency problem, which leads to the existence of the interest rate premium. In particular, after the purchase of the capital stock, each entrepreneur receives an idiosyncratic productivity shock ω that affects the capital holdings (ωK_{t+1}) and the return on capital holdings (ωR_{t+1}^K). Banks have to pay a “state verification” (monitoring) cost to infer the realized return. As a result, entrepreneurs have to pay an external finance premium over the riskless rate in order to borrow funds.

In the formal representation of the entrepreneurs' problem, we follow [Christiano et al. \(2010\)](#) and somewhat deviate from the original BGG specification, assuming that entrepreneurs are not directly involved in the production of intermediate goods. One of the implications of such a departure is that the net worth of entrepreneurs comes from profits (including capital gains) accumulated from previous capital investment.³

At the end of period t , entrepreneurs purchase capital K_{t+1} from capital-goods producers at price Q_t . Thus, the amount of borrowed funds is given by $B_{t+1} = Q_t K_{t+1} - N_{t+1}$. After observing the $t+1$ shock, the entrepreneur decides on the degree of capital utilization (U_{t+1}) and rents a part of the capital services to intermediate-goods firms at rate $\hat{r}_{t+1}^k$. A non-depreciated capital stock is then sold at price Q_{t+1} . The amount of effective capital that entrepreneurs can rent to firms is $K_{t+1}^S = U_{t+1} K_{t+1}$. The income from renting capital services is $\hat{r}_{t+1}^k U_{t+1} K_{t+1}$, while the (real) cost of changing capacity utilization is $a(U_{t+1}) K_{t+1}$, where a is a convex function with $a', a'' > 0$. The entrepreneur, who receives an idiosyncratic productivity shock ω , chooses U_{t+1} to solve:

$$\max_{U_{t+1}} \left[\hat{r}_{t+1}^k U_{t+1} - a(U_{t+1}) \right] \omega K_{t+1}. \quad (5)$$

The (aggregate) solution implies that

$$\hat{r}_{t+1}^k = a'(U_{t+1}). \quad (6)$$

Then optimal log-linear representation of capital utilization condition takes the form:

$$\hat{u}_t = ((1-\psi)/\psi) \hat{r}_t^k, \quad (7)$$

where ψ is the elasticity of the capital utilization cost function. The log linearized relation for capital services is given by

$$\hat{k}_{t+1}^S = \hat{u}_{t+1} + \hat{k}_{t+1}. \quad (8)$$

The average (aggregated over all the entrepreneurs) rate of return on capital purchased at time t is given by

$$E_t R_{t+1}^k = E_t \left[\frac{\hat{r}_{t+1}^k U_{t+1} - a(U_{t+1}) + Q_{t+1}(1-\tau)}{Q_t} \right], \quad (9)$$

where τ is the depreciation rate. Expression (9) equates the marginal return of capital, given by the right-hand-side terms, to the real expected interest rate on external funds. The log-linearized relation that describes the dynamics of the average

³ In the original BGG specification, the net worth also includes an income from labor, which entrepreneurs supply inelastically to the general labor market. However, [Bernanke et al. \(1999\)](#) note that the labor income source of the net worth, under reasonable parametrization, does not have a significant impact on the FA effects and the model dynamics.

expected real return to capital is given by

$$E_t \hat{R}_{t+1}^K = \frac{1-\tau}{\bar{R}^K} E_t \hat{Q}_{t+1} + \frac{\bar{r}^k}{\bar{R}^K} E_t \hat{r}_{t+1}^k - \hat{Q}_t, \quad (10)$$

where $\bar{R}^K$ denotes the steady-state return to capital and $\bar{r}^k$ is the steady-state rental rate.

The equilibrium condition on financial markets is derived from the optimal-debt contract problem, which maximizes the welfare of the entrepreneur, combined with the zero-profit condition of the bank. The details of the financial contract specification and derivations can be found in Appendix A of [Bernanke et al. \(1999\)](#). The optimality condition, which determines the link between the external financing costs, capital purchases and entrepreneurial financial position, is given by

$$E_t R_{t+1}^k = E_t \left[s \left(\frac{N_{t+1}}{Q_t K_{t+1}} \right) \varepsilon_t^b R_t \right]. \quad (11)$$

Eq. (11) indicates that the cost of external financing is composed of the premium for borrowed external funds represented by a function $s \left(\frac{N_{t+1}}{Q_t K_{t+1}} \right)$, the risk-free interest rate and an exogenous disturbance that describes fluctuations in the risk premium not captured by the financial frictions of [Bernanke et al. \(1999\)](#). Therefore, the FA mechanism is driven by the risk premium which consists of both endogenous and exogenous components. The exogenous shock is added to understand to what extent Bernanke-type financial frictions explain the dynamics of the risk premium⁴. In addition, the presence of the exogenous financial shock facilitates the comparison of models with and without a FA.⁵ The log-linearized condition (11) is represented by the following equation:

$$E_t R_{t+1}^k = -el \left\{ E_t [\hat{N}_{t+1} - \hat{Q}_t - \hat{k}_{t+1}] \right\} + \hat{R}_t + \hat{b}_t, \quad (12)$$

where $\hat{R}_t = (\hat{R}_t^n - E_t[\hat{\pi}_{t+1}])$ is the risk-free real interest rate and el represents the elasticity of the external finance premium to the change in the financial conditions. The equilibrium equation presented above implies that the capital will be purchased up to the point where the expected real return on capital is equal to the marginal cost of external finance. The higher the fraction of the project value financed by the entrepreneur's internal funds N relative to the gross value of capital QK , the lower the capital market friction and the lower the corresponding risk premium. It is easy to see that the model with a FA can be reduced to the [Smets and Wouters \(2007\)](#) model. In particular, the absence of financial frictions implies the case when entrepreneurs have sufficient net worth to finance the demand for capital stock. In such a situation, the risk of default associated with borrowing external funds vanishes, the risk-free rate and the real return to capital coincide. Thus, after setting the amount of borrowing and/or the parameter elasticity of the external finance premium to zero and combining Eqs. (10) and (12), we obtain a standard relation which determines the asset price dynamics in the [Smets and Wouters \(2007\)](#) model.

The law of motion for the aggregate financial wealth of entrepreneurs is given by

$$N_{t+1} = \chi \left[R_t^K Q_{t-1} K_t - E_{t-1} R_t^K (Q_{t-1} K_t - N_t) \right] + W_t^e, \quad (13)$$

where χ is the entrepreneurial survival rate and W_t^e is the transfer, which provides new entrant entrepreneurs who are in business in period t with the funds needed to buy some capital. Eq. (13) demonstrates that the aggregate net worth of surviving entrepreneurs is equal to the difference between the revenue from capital holding in time t and the cost of borrowing carried over from the previous period (the rate of interest paid by entrepreneurs on loan contracts B_t signed in time $t-1$), averaged across all the entrepreneurs. The log-linearization of Eq. (13) leads to the expression of entrepreneurial net worth in the form of the following accumulation equation:

$$\hat{N}_{t+1} = \chi \bar{R}^K \left[\frac{\bar{K}}{\bar{N}} (\hat{R}_t^K - E_{t-1} \hat{R}_t^K) + E_{t-1} \hat{R}_t^K + \hat{N}_t \right], \quad (14)$$

where $\bar{K}/\bar{N}$ is the steady-state ratio of capital to net worth, i.e. the inverse of the leverage ratio. Eq. (14) demonstrates that, in general terms, the endogenous variations in net worth in the next period come from unexpected changes in the real return on capital. These unexpected changes might come from the shocks that reduce the rental rate of capital or the market value of capital. As a result of such shocks, the entrepreneurial net worth will drop, leading to a fall in investment. As will be shown below, learning process may impact this traditional FA channel. In particular, both the rental rate and the price of capital are forward-looking variables, affected by expectations. Departure from the complete rationality assumption by allowing agents to formulate predictions that reflect their own perceptions about the economy can modify the evolution of financial variables. This, in turn, may have a substantial effect on the entrepreneurs' financial positions and a real economy.⁶

⁴ [Gilchrist et al. \(2009\)](#) interpret this financial disturbance as a shock to the supply of credit that captures the efficiency of the financial intermediation process or a shock to the financial sector that boosts the external finance premium beyond the level warranted by the current economic conditions.

⁵ [Smets and Wouters \(2007\)](#) model specification includes the exogenous equity premium shock which affects the price of credit.

⁶ As I will demonstrate, in this model, the variability of asset prices due to learning is one of the main sources of the additional volatility of the net worth and the finance premium, especially if firms are highly leveraged.

Combining Eqs. (12) and (14), the net worth can be expressed as a function of the risk-free interest rate and the exogenous and endogenous finance premium:

$$\hat{N}_{t+1} = \kappa \bar{R}^K \left[\frac{\bar{K}}{\bar{N}} \hat{R}_t^K - \left(\frac{\bar{K}}{\bar{N}} - 1 \right) (\hat{R}_{t-1} + \hat{b}_{t-1}) - el \left(\frac{\bar{K}}{\bar{N}} - 1 \right) (\hat{k}_t + \hat{Q}_{t-1} - \hat{N}_t) + \hat{N}_t \right]. \quad (15)$$

The values of the parameters κ , $\bar{K}/\bar{N}$ and el determine the impact of financial frictions on the real economy. The higher the entrepreneurial survival rate and the capital to the net worth steady-state ratio, the more persistent the evolution of net worth will be. Combined with the higher elasticity of the external finance premium, this would imply a stronger response of the wedge between the expected return on capital and the risk-free rate. Therefore, shocks affecting entrepreneurial net worth would have greater real effects.

2.2. Monetary policy and equilibrium

The model is completed by adding the following empirical monetary policy reaction function:

$$\begin{aligned} \hat{R}_t^n = & \rho_R \hat{R}_{t-1}^n + (1 - \rho_R)(r_\pi \hat{\pi}_t + r_y \widehat{ygap}_t) \\ & + r_{\Delta y}(\widehat{ygap}_t - \widehat{ygap}_{t-1}) + \epsilon_{r_t}. \end{aligned} \quad (16)$$

The monetary authority follows a generalized Taylor rule responding to inflation and the output gap terms (current and lagged). The latter is defined as the difference between actual and potential output. The output gap is approximated by $\widehat{ygap}_t = \hat{y}_t - \hat{A}_t$. The parameter ρ_R captures the degree of interest rate smoothing. ϵ_{r_t} is an iid-Normal monetary policy shock.

The log-linear representation of the resource constraint is given by

$$\hat{y}_t = \frac{(\bar{R}^K - 1 + \tau)k_*}{y_*} \hat{u}_t + \hat{\mu}_t^{bank} + \frac{c_*}{y_*} \hat{c}_t + \frac{i_*}{y_*} \hat{i}_t + \hat{g}_t, \quad (17)$$

where $\hat{\mu}_t^{bank} = (k_*/y_*)(\bar{R}^K - \bar{R})(1 - \bar{N}/\bar{K})(\hat{R}_t^K + \hat{Q}_{t-1} + \hat{k}_t)$ measures the bank monitoring cost; $\hat{g}_t$ is exogenous government spending, which is assumed to follow a first-order autoregressive process with an iid-normal error term and is also affected by the productivity shock as in [Smets and Wouters \(2007\)](#): $\hat{g}_t = \rho_g \hat{g}_{t-1} + \rho_{ga} \epsilon_t^a + \epsilon_t^g$. The relation between domestic productivity and government spending is motivated by the fact that, in estimation, the exogenous spending component also includes net exports, which may be affected by domestic productivity developments.

2.3. Introducing adaptive learning

I depart from the assumption of RE and assume that agents possess incomplete knowledge about the economic environment (model structure and parameters). Since agents do not know the Rational Expectation Equilibrium (REE), they are unable to produce model-based predictions of the path of forward-looking variables and have to form their own beliefs about future developments on the basis of the information they observe. As in [Marcet and Sargent \(1989\)](#) and [Evans and Honkapohja \(2001\)](#), agents gradually learn the “true” parameters of the model by adapting their expectations with the use of a certain learning algorithm. In this section, I present a general description of a Kalman filter learning setup.⁷ For more details on the description and implementation of this learning process, see [Slobodyan and Wouters \(2012b\)](#).

The model described in [Section 2](#) can be represented in the following structural form:

$$A_0 \begin{bmatrix} y_{t-1} \\ w_{t-1} \end{bmatrix} + A_1 \begin{bmatrix} y_t \\ w_t \end{bmatrix} + A_2 E_t y_{t+1} + B_0 \epsilon_t = \text{const.}, \quad (18)$$

where the vector y_t includes endogenous variables of the model and w_t is an exogenous AR(1) process:

$$w_t = \Gamma w_{t-1} + \Pi \epsilon_t. \quad (19)$$

In general form, the RE solution of the model (18) is given by the following expression:

$$\begin{bmatrix} y_t \\ w_t \end{bmatrix} = \mu + T \begin{bmatrix} y_{t-1} \\ w_{t-1} \end{bmatrix} + R \epsilon_t, \quad (20)$$

where T and R are time-invariant non-linear functions of the model structural parameters Θ . Under RE, the intercept μ is normally a vector of zeros.⁸ In my case, vector y_t contains 16 endogenous variables. In addition, the stochastic structure, represented by the vector w_t and innovations ϵ_t , is determined by seven exogenous shocks. The model variables

⁷ The alternative, widely used, learning method is the constant gain Recursive Least Squares (RLS). [Sargent and Williams \(2005\)](#) demonstrate that both learning methods mentioned above are asymptotically equivalent on average. However, their transitory behavior may differ significantly. In particular, the Kalman filter tends to result in a faster adjustment of agents' beliefs due to the fact that the learning gain is not assumed to be constant.

⁸ It can be a nonzero vector only for observable variables that are not detrended.

summarized by vector y can be grouped into the state variables y^s that appear with a lag, forward variables y^f that appear with a lead, and the so-called static variables.⁹ Agents have to forecast seven forward-looking variables, such as: consumption, investment, hours worked, price and wage inflations, return on capital and asset prices. Deviation from the RE assumption implies that agents do not have sufficient information about the model and its parameters to formulate the REE-based expectations $E_t y_{t+1}$. Thus, they cannot derive the law of motion (20). To obtain the solution, they have to formulate the so-called Perceived Law of Motion (PLM), which relates the value of the forward variable j to the model state variables X_j using a reduced-form linear function:

$$y_{j,t}^f = \beta_{j,t-1} X_{j,t-1} + u_{j,t}. \quad (21)$$

The forecasting model is then represented in the SURE format and composed of the variable-specific equations of the form (21). In this paper, in period t , agents predict only next-period variables which are present in Euler equations of firms and consumers. Thus, I implement the *Euler equation learning*, promoted by Evans and Honkapohja (2001).¹⁰ The error term $u_{j,t}$ in (21) consists of linear combination of the true model errors ϵ_t and has a variance–covariance matrix Σ . Matrix X_j includes a set of variables that are used to form predictions about forward-looking variable j . Thus, X_j may consist of all the state variables of the model y^s or a subset of y^s . In this paper, the model with learning is estimated using three alternative specifications of information set X_j , which will be discussed in greater details in the Estimation section. Note, that the functional form of (21) generally corresponds to the predictions obtained under RE. What learning brings into the set-up is the time variation in β_j due to periodical updates of beliefs, and the possibility for the coefficients β_j to diverge from the values implied by the RE solution (20). In addition, the vector of beliefs β_j may also include the constant which would mean that agents make inferences not only about the persistence of economic processes but also about growth rates. Finally, the information set X_j can be significantly more restrictive than the vector of all the state variables y^s , which would be used under RE.

The Kalman filter learning algorithm consists of the following steps:

1. *Formulation of the initial beliefs*: Following Slobodyan and Wouters (2012a) and Slobodyan and Wouters (2012b), we apply the common assumption in the learning literature and assume that initial beliefs $\beta_{1|0}$ are consistent with the REE.¹¹ Thus, the initial values for the vector of beliefs β and variance–covariance matrix Σ , needed to initialize the Kalman filter, are based on the matrix of second moments Ω of the model variables, which is derived for the REE given by (20) and evaluated for the parameter vector θ . The moments imply a relationship between the forward-looking variables y^f and the variables X used in forecasting equations. Then, using the corresponding rows and columns of Ω , $\beta_{1|0}$ is obtained as the projection of X on y^{12} :

$$\beta_{1|0} = E[X'X]^{-1} \cdot E[X'y],$$

where the expectations $E[\cdot]$ are taken from the RE solution. In the estimation procedure, initial beliefs always correspond to the REE for the currently evaluated (in the posterior optimization or MCMC steps) parameter vector.¹³ Therefore, this approach of choosing the initial beliefs is the closest to the pure rational expectations. Given $\beta_{1|0}$, the variance–covariance matrix is calculated as:

$$\Sigma = E \left[\left(y_t^f - X_{t-1} \beta_{1|0} \right) \left(y_t^f - X_{t-1} \beta_{1|0} \right)' \right].$$

2. *Update*: The Kalman filter is used to obtain the updated estimates of the vector of beliefs β and the error covariance matrix P :

$$\beta_{t/t} = \beta_{t/t-1} + K_t \tilde{z}_t, \quad (22)$$

$$P_{t/t} = (I - K_t X_{t-1}) P_{t/t-1},$$

where the innovation or measurement residual $\tilde{z}_t = y_t^f - \beta_{t/t-1}' X_{t-1}$, the innovation (or residual) covariance $S_t = \Sigma + X_{t-1}' P_{t/t-1} X_{t-1}$; and the optimal Kalman gain $K_t = P_{t/t-1} X_{t-1}' S_t^{-1}$. Updating of the beliefs at any t depends on the forecast error, data (best estimates of the variables at time $t-1$) and on the initial beliefs.

3. *Prediction*: According to agents' beliefs, the coefficients β follow a vector autoregressive process:

$$(\beta_t - \bar{\beta}) = F \cdot (\beta_{t-1} - \bar{\beta}) + v_t, \quad (23)$$

⁹ y^f and y^s could intersect.

¹⁰ An alternative type of learning—long-horizon learning, advocated by Preston (2005)—implies that agents forecast economic variables infinitely many periods ahead.

¹¹ Slobodyan and Wouters (2012a) discuss alternative ways of evaluating the initial beliefs, which may significantly deviate from the REE. Specifically, initial beliefs can be constructed from the pre-sample regressions or derived on the basis of optimized posterior probability as in Sargent et al. (2006). Slobodyan and Wouters find that dynamics under the learning process can be very sensitive to initial beliefs. In the situation when agents' beliefs significantly deviate from the beliefs implied by the REE, model dynamics under learning can be driven rather by transition from the initial beliefs to the equilibrium beliefs than by the standard updating due to new information.

¹² We use the fact that estimator β_{OLS} is unbiased.

¹³ In this respect, initial beliefs are called to be “model-consistent”.

where F is a diagonal matrix with $\rho \leq 1$ on the main diagonal, $\bar{\beta}$ is the the initial beliefs and v_t are i.i.d. errors with variance–covariance matrix V . Parameter ρ , which is assumed to be the same for all the seven forecast variables, measures the intensity of updates and therefore is referred to the learning “gain” parameter, which is estimated. Using (23), the forecast of the evolution of β can be obtained as follows: $(\beta_{t+1/t} - \bar{\beta}) = F \cdot (\beta_{t/t} - \bar{\beta})$. The predicted estimate covariance is given as: $P_{t+1/t} = F P_{t/t} F' + V$. The variance–covariance matrix V of shocks v_t and an initial guess for P_{10} are taken to be proportional to the GLS estimator of the variance of β^{14} : $V = \sigma(X' \Sigma^{-1} X)^{-1}$ and $P_{10} = \gamma(X' \Sigma^{-1} X)^{-1}$, where scaling parameters σ and γ can be calibrated or estimated.

The best estimates of the beliefs $\beta_{t/t-1}$ generated in the prediction step of the Kalman filter are then used to calculate expectations $E_t y_{t+1}$ of forward-looking variables according to forecasting Eqs. (21). Substituting these expectations into the structural representation of the model (18), we obtain the Actual Law of Motion (ALM) of the system in a purely backward-looking form:

$$\begin{bmatrix} y_t \\ w_t \end{bmatrix} = \mu_t + T_t \begin{bmatrix} y_{t-1} \\ w_{t-1} \end{bmatrix} + R_t \epsilon_t. \quad (24)$$

Introducing AL does not affect the initial steady state of the system, i.e. at time $t=0$ we start from the RE equilibrium solution given by Eq. (20).

In general terms, the transmission under learning can be summarized as follows. Following the innovations in exogenous disturbances and subsequent changes in the variables included in the information set, agents start to adjust their belief coefficients iterating over Eq. (22). In expressing their perceptions about the impact of the shock on the economy, they have two instruments in their disposal: to update the autoregressive coefficients, which determine the persistence of the forecast variables, and the mean of the variable. Since the effect of the learning process operates through expectations, equations that define the intertemporal decisions of agents and thus directly involve the expectation terms¹⁵ become particularly important for transmitting the impact of the shocks to changes in the belief coefficients and afterwards onto the estimated structural parameters and the model dynamics. In particular, the forward-looking equations that largely determine the dynamics under learning are the Euler equation for consumption, inflation Phillips curve, investment equation and one of the key FA Eqs. (10), which describes the evolution of the average expected real return on capital. For example, exogenous shock to the risk premium will mostly impact expectations (and beliefs) about the real return on capital and asset prices via Eqs. (10) and (12). In particular, following the positive risk premium shock, asset prices will show a decline. Such an observed downturn on financial markets motivates agents to revise perceptions about asset price persistence upwards. At the same time, the mean can be adjusted downwards if agents believe that the shock will have a rather pronounced negative effect on the growth rate of asset prices. These beliefs will be reflected in the expected trend in asset prices through the forecasting Eq. (21) and will feed into the dynamics of the endogenous variables through the ALM (24). Therefore, the simulation of the system's dynamics under learning entails calculation of a time-varying transmission mechanism determined by μ_t , T_t and R_t , which are the functions of the model structural parameters given by matrices A_0, A_1, A_2 and B_0 as well as beliefs $\beta_{t/t-1}$. The intensity of time-variation determines to what extent the equilibrium path under learning differs from the RE equilibrium. In our case, when agents' beliefs are chosen to be consistent with REE, we expect that most of the updates will be driven by adjustments of beliefs due to new information.

Note that the specification of forecasting equations implies that agents do not directly observe and distinguish between exogenous processes. Learning agents make inferences only on the basis of the observed variables affected by the shock (and the available information set can be very limited). As a result, the effects of certain shocks on the equilibrium dynamics under learning can be over-estimated or, on the contrary, realized more slowly and incompletely compared to the transmission under RE. The transmission speed depends to a significant extent on the nature of the shock. In particular, in forecasting inflation, productivity or price mark-up shocks, which have a direct impact on the marginal costs (and thus on the inflation Phillips curve relation), are realized quicker compared for example to the monetary policy shock, which is transmitted into prices via its impact on investment and demand.¹⁶

¹⁴ β_{GLS} is an efficient estimator for the SURE model.

¹⁵ Parameters of the matrix A_2 in (18) have non-zero values.

¹⁶ Slobodyan and Wouters (2012b) demonstrate that learning agents tend to overestimate the effect of price mark up shock. As a result, the inflation response is more persistent, which in turn determines a less aggressive monetary policy reaction and a lower decline in the aggregate demand. In this respect, the ability of learning to generate more gradual responses of certain variables to shocks makes the dynamics under learning more realistic and consistent with the responses produced by VAR and DSGE-VAR models.

3. Estimation strategy and results

3.1. Data and measurement equations

The model is estimated using seven macroeconomic quarterly U.S. time series: real GDP, real consumption, real investment, real wage, hours worked, GDP deflator and the federal funds rate. Nominal variables are deflated by the GDP deflator. Aggregate variables are expressed in per capita terms. All variables except hours, inflation, and interest rates are taken in first differences. Financial variables are not included in the set of observables in order to facilitate the comparison with [Smets and Wouters \(2007\)](#) and [Slobodyan and Wouters \(2012b\)](#) and other previous studies¹⁷. I estimate the model for the sample period 1954:1–2008:3. The long data sample is chosen in order to assess the importance of time variation in the model parameters introduced by AL. The sample ends in 2008 because the (linear) model framework is not designed to capture the impact of non-linearities, caused by the interest rate zero lower bound and unconventional policy measures implemented during that period. The link between the model variables and the data is given by a set of the following equations:

$$\begin{bmatrix} dlGdp_t \\ dlCons_t \\ dlInv_t \\ dlWage_t \\ lHours_t \\ dlP_t \\ FedFundsR_t \end{bmatrix} = \begin{bmatrix} \bar{\gamma}_y \\ \bar{\gamma}_c \\ \bar{\gamma}_i \\ \bar{\gamma}_w \\ \bar{l} \\ \bar{\pi} \\ \bar{r} \end{bmatrix} + \begin{bmatrix} \hat{y}_t - \hat{y}_{t-1} \\ \hat{c}_t - \hat{c}_{t-1} \\ \hat{i}_t - \hat{i}_{t-1} \\ \hat{w}_t - \hat{w}_{t-1} \\ \hat{l}_t \\ \hat{\pi}_t \\ \hat{R}_t^n \end{bmatrix}, \quad (25)$$

where l and dl stand for log and log difference, respectively. Unlike [Smets and Wouters \(2007\)](#), I estimate separately the trends for output, consumption, investment and wages growth rates, instead of imposing a common trend on these variables. $\bar{\pi} = 100(\Pi_* - 1)$ is the quarterly steady-state inflation rate and $\bar{r} = 100(\bar{\gamma}^{\sigma_c} \Pi_* / \beta - 1)$ is the steady-state nominal interest rate. Given the estimates of the average trend growth rate and the steady-state inflation rate, the latter will be determined by the estimated discount rate. Finally, $\bar{l}$ refers to steady-state hours worked.

3.2. Bayesian estimation under adaptive learning

I estimate several model specifications, and in particular, versions with and without financial frictions in order to assess the empirical validity of the FA mechanism. In addition, I estimate each model under the assumption of RE and with learning. So, I have two dimensions of comparison—the effect of financial frictions and the impact of expectations. When assessing the effects of the FA under learning, I experiment with alternative AL schemes, which differ in terms of the variables used by agents to form forecasts. The log-linearized versions of the models are estimated using Bayesian methods.¹⁸ Typically, from 300,000 to 500,000 MCMC draws were performed, using three chains.¹⁹

For estimation purposes, the model is represented in the state-space form, which combines Eq. (24), the so-called state equation, with the measurement Eqs. (25). Given the state space representation of the model, the Kalman filter is used to evaluate the likelihood. For evaluating the model under imperfectly rational beliefs, I use the toolbox developed by [Slobodyan and Wouters \(2012b\)](#), which implements procedures for AL within the Dynare 3.064 Matlab toolbox. As described in Section 2, the estimation procedure under learning differs from the inferences under the RE due to time variation in beliefs (22), which implies the dynamic solution of μ_t , T_t and R_t ²⁰. The values of μ_t , T_t and R_t are used to form expectations of the next period model variables in the main Kalman filter step and are used to calculate the model likelihood. In other words, the standard Kalman filter incorporates an additional filtering stage used to update agents' beliefs (see Section 2.3). It is important to note that endogenous model variables used on both sides of the forecasting Eqs. (21) are not, in fact, observed.²¹ Therefore, we use the model counterparts (more specifically, filtered estimates $\hat{y}_{t/t}^f$ and $\hat{y}_{t-1/t-1}^s$) produced by the Kalman filter. The current learning procedure does not employ smoothed estimates of the variables, which would imply more efficient use of the available information by allowing for the possibility of revisions of past filtered estimates and beliefs.

The time-varying procedure makes T_t a complicated function of the data, current parameters, and beliefs that could easily become unstable for one or several periods. Such discontinuities in changes in beliefs lead to numerical problems during estimation and lower-quality estimation results. In particular, allowing T_t to be explosive for some periods leads to an

¹⁷ A number of studies, which compare DSGE models with and without FA, such as [De Graeve \(2008\)](#) and [Christensen and Dib \(2008\)](#) also perform the estimation without including financial variables into the data set.

¹⁸ For more details on Bayesian estimation of DGSE models, see [An and Schorfheide \(2007\)](#).

¹⁹ For the small scale AL models (AR(2) and AR(2) + constant) with the slower convergence, I checked the results up to 3 000 000 draws.

²⁰ These matrices do not change over time during the estimation under RE.

²¹ In the state-space representation of the model, endogenous variables and the observed variables used to estimate the model are linked through the measurement equations.

increase in forecasting errors and thus to a much worse likelihood. In this paper, I have to deal with explosive dynamics of T_t for some time periods when estimating the DSGE model under non-MSV learning (when agents use a limited information set to form predictions). This problem seems to be more serious for estimations of a model with a FA. In particular, financial frictions introduce additional volatility, which in turn may lead to more frequent and sizeable adjustments of beliefs. Thus the probability of the eigenvalues of T_t to jump outside the unit circle also increases. However, in all the estimations I performed, the number of periods with unstable eigenvalues does not exceed 5. As is common in the learning literature, I use a projection facility that skips updating in such cases.

In this paper, I estimate several model specifications, which differ in terms of information sets (given by matrix X_j in (21)) used by agents to form their beliefs about the forward-looking variables:

- “AR(2)+constant”: the forecasting Eq. (21) for every forward-looking variable is a simple univariate AR model with two lags and a constant.²² Thus, agents form and update their beliefs about the persistence and the mean of the variables.
- “AR(2)”: the forecasting equation for every forward-looking variable includes only two own lags.
- “All states”: the forecasting equation for every forward-looking variable includes all the state variables. In this case, the information set X_j corresponds to the state vector used under RE. Thus, the functional form of the relationship between forward and state variables is very similar to the MSV REE reduced form.

The priors are chosen following [Smets and Wouters \(2003, 2007\)](#). These papers present a detailed description of the estimation methodology as well as the justification for the choice of priors. The priors for additional parameters related to the financial frictions are based on calibration exercises and previous literature ([Bernanke et al., 1999](#) and [De Graeve, 2008](#)). In particular, $\bar{R}^K \sim Normal(1.0149, 0.002)$. α , $\bar{k}/\bar{N}$, and el are assumed to have Uniform priors with sufficient standard deviations. The choice of the flat and disperse priors makes it possible to check whether macroeconomic time series used in the estimation actually contain information about financial frictions parameters.

3.3. Estimation results. model fit

[Table 1](#) reports the logarithms of marginal data densities for the various estimated specifications. I compare the results for models with RE and AL. In addition, each version was estimated with the financial accelerator mechanism (FA) and without (noFA).

The estimation results suggest that the model with RE and FA fits the data much better than the version that does not incorporate financial frictions (similar to the findings of [De Graeve, 2008](#) and [Christensen and Dib, 2008](#)). In addition, as in [Slobodyan and Wouters \(2012b\)](#), the ability of AL to improve the data fit can be clearly observed for “noFA” specifications. In particular, [Table 1](#) indicates that relaxing the RE hypothesis through the introduction of a Kalman filter AR(2) learning significantly improves the fit of the model. The performance of the AL model based on more complicated forecasting functions (i.e. “all states”) is essentially the same as the RE model. [Table 1](#) also demonstrates that the improvement in the data fit under learning is lower once financial frictions are introduced. In order to shed more light on the estimation results, I analyze the relative likelihood (evaluated at the posterior mode) of alternative model specifications as a function of time. I would like to find out how introducing AL changes the empirical performance of the model over time relative to the model with RE. In other words, it is important to understand whether the transmission mechanism of the best-performing learning model can be more suitable for describing the macro-dynamics during certain economic episodes. Firstly, I will illustrate graphically how the departure from the RE hypothesis affects the data fit in models with and without a FA. [Fig. 1](#) shows the cumulative likelihood for “AR(2)+constant” learning models with and without a FA relative to the corresponding RE models.²³ The upward trend of the cumulative difference line indicates that on average the likelihood of the learning model on this time interval is better relative to the likelihood implied by the model based on RE.

[Fig. 1](#) indicates that the RE and learning models estimated with or without a FA deliver very similar relative data fits before the mid 1970s. In my set-up, such a result might imply that financial disturbances did not play a significant role in the evolution of the real economy. In addition, on average the RE model fits the data better than the learning model. Therefore, it appears that before the middle of the 1970s other shocks than those which impact and propagate through the expectations dominated in driving the macroeconomic dynamics. In particular, we could think about fiscal shocks, which do not directly affect the intertemporal choices of agents.²⁴

[Fig. 1](#) also demonstrates that the aptitude of the learning model to describe the data-generating process has been improving slowly but surely since the late 1980s. This could imply that shocks affecting the expectations (price and wage mark-up shocks, as well as financial shocks) became more important in driving the macroeconomic dynamics.

The graphs presented in [Fig. 1](#) also demonstrate that the relative performance of the RE and learning models in “FA” and “noFA” specifications show different trends during the middle 1970s and 1980s. This period was characterized by two severe

²² Parsimonious AR(2) form of forecasting equations has been chosen in order to ensure simple structure and interpretation. I have also experimented with AR(1) forecasting equations, which showed inferior performance comparing to AR(2) in terms of the model fit.

²³ I compute the difference in likelihood for AL and RE models, and plot the cumulative sum of this difference.

²⁴ In this paper, I do not aim at explaining the sources of macroeconomic fluctuations by offering a detailed analysis of the shock decomposition. I rather attempt to add to understanding of the factors that could explain the performance of AL during certain time periods.

Table 1

Model comparison in terms of marginal likelihood.

Model specification	FA	noFA
RE	– 1207.8	– 1231.52
Kalman Filter AL		
AR(2)+constant	– 1198.39	– 1209.8
AR(2)	– 1201.44	– 1212.46
All states (near MSV)	– 1206.11	– 1234.55

Notes: Log marginal data densities for the models with and without FA. Learning specifications vary depending on the information set used to form predictions; initial beliefs are REE-consistent.

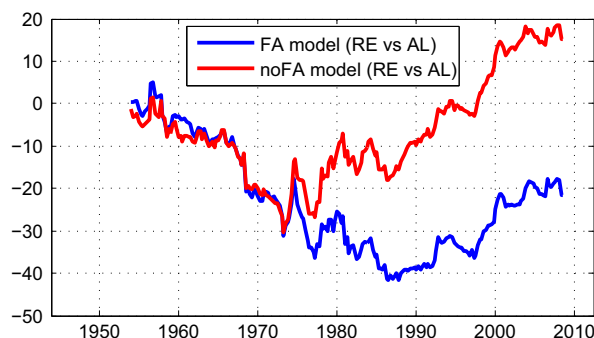


Fig. 1. Cumulative likelihood for the “AR(2)+const.” learning specification with and without a FA relative to the corresponding RE model.

recessions accompanied by an increased volatility of inflation, consumption, investment growth and hours. In the version without a FA, the time variation introduced by AL enables capturing the dynamic changes in public's expectations and thus contributes to explaining the increased instability. In other words, learning can partially substitute for the absence of financial market inefficiencies. The inclusion of financial frictions considerably improves the performance of the RE model, especially during the mid 1970s and the beginning of the 1980s (see also Fig. 2, which compares the cumulative likelihood for the RE and “AR(2)+const.” learning models, in the set ups with and without FA). Thus, the relative gain from modeling the time-varying transmission in a model with FA is somewhat reduced²⁵.

Model fit measured by a marginal likelihood is closely related to the one-period-ahead prediction error. Fig. 3 compares the one-step-ahead (in sample) forecasting performance of the RE and the best performing AL model.²⁶ In particular, I contrast the forecasts and the actual dynamics of inflation and investment growth. The forecasts are computed up to the 2011q3. The analysis of the model performance on the data interval covering the recent financial crisis is of particular interest.²⁷ Section A.1 of Appendix A presents the RMSE statistics.

As described in Section 2.3, forecasting models imply that agents have very limited information set and cannot discriminate between the shocks that hit the economy in a specific moment of time. Admittedly, such an assumption cannot be considered as very realistic, but allowing agents to know and/or learn the structural shocks would significantly complicate the estimation procedure. A simple form of forecasting equations may facilitate or deteriorate the forecasting performance of AL comparing to the model with RE. In particular, rational agents have the possibility to anticipate future policy reactions and incorporate this knowledge into the predictions. In such cases, learning may react quicker and be more successful in predicting the abrupt changes in the certain variables. On the other hand, learning agents, who observe only two lags of the forecast variables, might be unable to identify the shock in time and therefore could react with a certain delay. As a result, the overall economic impact might be underestimated.

Let us consider several economic episodes during which AL showed better and worse forecasting performance relative to the model with RE.

²⁵ In the next subsections I will compare the implied persistence of the main macroeconomic variables under assumptions RE and AL and will add to explaining the results presented above.

²⁶ In this paper, I focus on one-quarter-ahead forecasts, which is the measure of the predictions consistent with the Euler equation learning approach. Slobodyan and Wouters (2012b) demonstrate that forecast performance of the AL model generally deteriorates at longer horizons. I leave for future research the comparison of the ability of the RE and AL models with financial accelerator to forecast $t+k$ periods ahead.

²⁷ The period 2008q4–2011q3 is not included in the baseline estimation sample due to the fact that “crisis” observations can distort the estimation results. Therefore, in order to check the ability of the RE and AL models to forecast the most recent economic downturn, I fix the parameters at the corresponding posterior mode values and run the forecast forward without a systematic update.

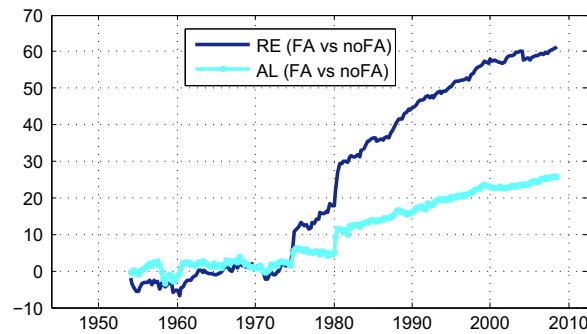


Fig. 2. Cumulative likelihood for RE and AL models with “FA” relative to the corresponding “noFA” models.

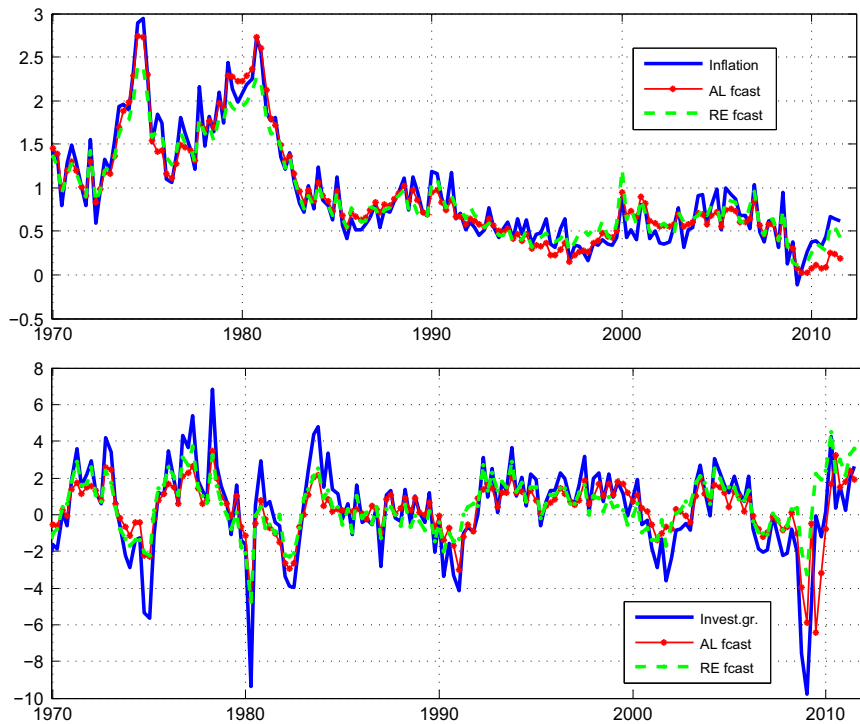


Fig. 3. One-step-ahead forecasting performance under RE and AL.

Fig. 3 illustrates that the AL model was more successful in predicting inflation in the middle of the 1970s and the beginning of the 1980s (see also Fig. 1, which shows an improvement in relative likelihood during these periods). Slobodyan and Wouters (2012b) explain this result by more realistic modeling of private sector inflation expectations. More specifically, they show that inflation persistence implied by the learning model shows significant time variation (increase in 1970s and 1980s) due to the impact of price and wage markup shocks, which were active during this time period.

The AL model did better in predicting the decline in output and investment growth during the recessions in the beginning of the 1980s and 1990s as well as the most recent economic downturn, characterized by a particularly pronounced drop in the real sector. In my set up, the departure from the RE hypothesis in modeling the asset price expectations plays a significant role in explaining this result. In particular, the latest financial distress was driven by a shock originated in financial sector. The disturbance was directly transmitted into financial variables and asset prices in particular leading to the sizable and fast updates of the asset price expectations. As a result, agents perceived asset prices as being more persistent, which resulted in the more pronounced real effects implied by AL model comparing to the model with RE.

On the other hand, AL was less successful in predicting the economic downturn during the recession of 2001 (see also Figs. 1 and 2). Such a result can be explained by the fact that the information set available to the learning agents appeared to be too restrictive for the proper identification of this recession. In particular, recession of 2001 was unusual in several respects. First of all, in contrast to the typical downturn, consumption expenditure remained relatively stable. In addition, this period was characterized by an unexpectedly large increase in labor productivity and wages. Such a stable dynamics of

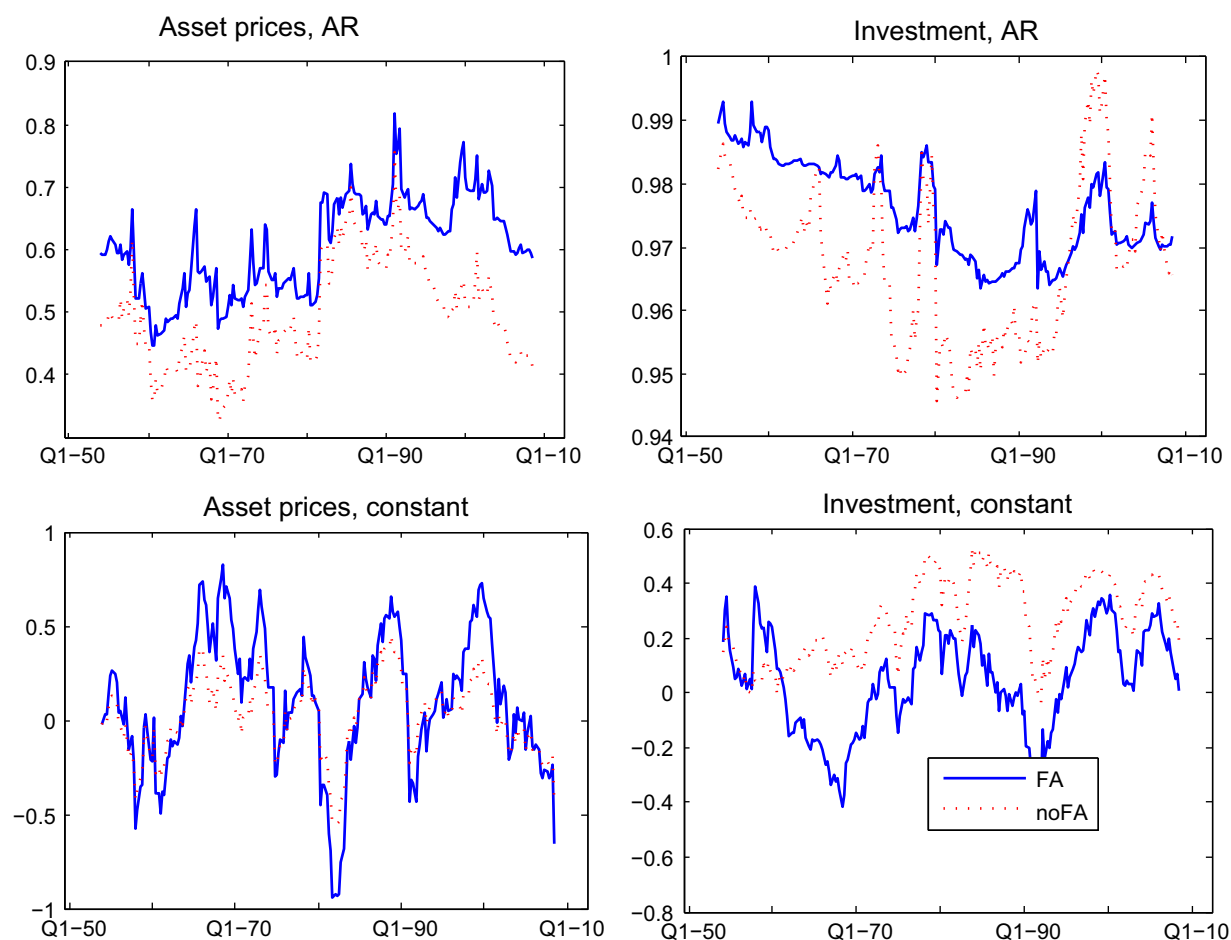


Fig. 4. Beliefs about asset prices and investment: Persistence and constant.

these important forward-looking variables were reflected in rather optimistic agents' expectations and could act as an offsetting factor to the negative shocks affecting investment and output. Secondly, the 2001 recession brought a particularly strong decline in investment. In forecasting such a dynamics, the performance of AL is, first of all, determined by ability of agents to identify an investment-specific shock from the recent trend in investment. Following the shock, agents did react to an investment decline by adjusting their beliefs (see also Fig. 4). In particular, the autoregressive coefficients in the forecasting equation for investment showed an increase, but learning agents estimated the persistence of the investment process at the lower level comparing to the agents with RE. In addition, perceptions about growth rates (the constant) were adjusted downwards, but only to a relatively small extent (compared to the previous recession in the 1990s). Therefore, it appears that AL agents did not perceive this period as being particularly severe for the real economy.

The main results of this section can be summarized as follows. Introducing both AL and the FA mechanism into the otherwise standard DSGE model can contribute to capturing the properties of real data over certain time intervals, especially in the second half of the data sample. Among the main factors determining performance of AL are the nature of shock affecting the economy and the speed of transition of the innovations to agents' expectations.

4. Financial frictions under learning. Time variation and transmission mechanism

AL can affect model fit in several ways. First, the time variation of beliefs allows the model to become time-varying (24). This could improve the empirical performance of the model if the process that generates the time series of the observed variables is itself time-varying. On the other hand, if the belief-updating process is too volatile, the uncertainty factor could lead to a deterioration of the fit. Another channel through which AL operates is through a change in the information set available to the agents. Even when beliefs are consistent with a REE and are not time-varying, the changes in the information set used by the agents to form the expectations will lead to a divergence of the transmission mechanism from that

Table 2

Comparison of RE and AR(2) learning models in terms of the estimated parameters.

Parameters		Prior distribution			Posterior, RE model			Posterior, AL model		
		Type	Mean	St.err	5%	Mean	95%	5%	Mean	95%
<i>Shocks</i>										
Exo.risk prem.	σ_b	I.Gam	0.1	2	2.3479	2.9965	3.5874	1.4758	2.0498	2.7821
Investment	σ_q	I.Gam	0.1	2	0.4536	0.5396	0.6221	0.3752	0.4309	0.4972
Price markup	σ_p	I.Gam	0.1	2	0.1344	0.1598	0.1844	0.1654	0.1849	0.2054
Wage markup	σ_w	I.Gam	0.1	2	0.1962	0.2245	0.2547	0.2042	0.2269	0.2476
<i>AR.coeff-s</i>										
Exo.risk prem.	ρ_b	Beta	0.5	0.2	0.0288	0.1253	0.2174	0.2175	0.3243	0.4441
Investment	ρ_q	Beta	0.5	0.2	0.5812	0.6709	0.7615	0.3003	0.4030	0.5272
Price markup	ρ_p	Beta	0.5	0.2	0.8372	0.9002	0.9634	0.0357	0.2018	0.3522
Wage markup	ρ_w	Beta	0.5	0.2	0.8985	0.9315	0.9699	0.2964	0.5279	0.7721
Pr.markup,ma	μ_p	Beta	0.5	0.2	0.6458	0.7752	0.9020	0.2763	0.4260	0.5637
W.markup,ma	μ_w	Beta	0.5	0.2	0.7944	0.8566	0.9303	0.1551	0.4146	0.6833
<i>Str.params</i>										
Adjust. cost	φ	Norm	4	1.5	4.7261	6.3475	7.9322	4.3979	5.9552	7.4496
Habit	η	Beta	0.7	0.1	0.6258	0.6873	0.7510	0.6850	0.7335	0.7967
Calvo wages	ξ_w	Beta	0.5	0.1	0.6737	0.7633	0.8500	0.6974	0.7500	0.8043
Calvo prices	ξ_p	Beta	0.5	0.1	0.6255	0.6914	0.7579	0.5259	0.5904	0.6455
Index. wages	ι_w	Beta	0.5	0.15	0.2534	0.4582	0.6378	0.0921	0.1914	0.2912
Index. prices	ι_p	Beta	0.5	0.15	0.0772	0.1933	0.3052	0.2382	0.4139	0.5849
Int.rate smooth	ρ_r	Beta	0.75	0.1	0.8352	0.8603	0.8859	0.8982	0.9193	0.9392
Pol rule, inflat.	r_π	Norm	1.5	0.25	1.6977	1.9354	2.1733	1.4630	1.7521	2.0222
Cap./net worth	k/N	U	1	3.5	2.3413	2.8907	3.5000	2.4914	3.0231	3.5000
Survival rate	κ	U	0.9	1	0.9142	0.9451	0.9787	0.9039	0.9466	0.9672
Elast.risk prem	ϵ_l	U	0	0.5	0.0031	0.0186	0.0318	0.0039	0.0196	0.0351
AL param.	ρ	Beta	0.5	0.289	–	–	–	0.9751	0.9842	0.9931

under RE. In this section, I will further examine how the expectation formation mechanism introduced by learning modifies the properties of the model and, as a result, affects the transmission of various shocks.

4.1. Parameter estimates

In this subsection, I discuss the impact of the financial frictions and the adaptively formed expectations on the estimated parameters. I present the MCMC optimization results for the RE and “AR(2)+constant” learning models estimated with and without financial frictions. I will also report the findings from the posterior maximization of all learning models and contrast them with RE. The results of the model comparison in terms of the estimated parameters are presented in [Tables 2 and 3](#).

I can see from the estimation results that there are important changes in the estimated structural parameters and the stochastic processes depending on the assumed expectation formation mechanism (RE or several forms of learning). Naturally, the most pronounced impact of learning will be on the parameters that enter the expectation equations such as the consumption Euler equation, inflation and wage Phillips curves, as well as the equations that determine the average return to capital and the external finance premium.²⁸

In particular, I find that some of the estimated structural rigidities and shock persistence decrease. More specifically, autoregressive components of exogenous processes for price and wage mark-up shocks fall significantly. The decline in the persistence of the shock to investment technology is not so dramatic, but still notable. The confidence interval for this parameter clearly shift left and do not overlap with the range of possible values implied by the posterior distribution of the RE model. In addition, the variance of the investment shock declines under learning. These results imply that learning is helpful in explaining inflation, wage and investment dynamics. Modeling adaptive expectations of these variables introduces “endogenous” persistence, which has an empirically appealing economic interpretation. At the same time, the autoregressive coefficient of the exogenous equity premium shock, which can be viewed as a proxy for financial-market inefficiencies, tends to increase. The variability of this shock declines. Thus, it appears that the time-varying transmission mechanism can account for some of the variation in the external finance premium, but cannot generate sufficient endogenous persistence in the financial variables through evolution of agents’ beliefs.

The degree of price rigidities, wage indexation and to some extent wage stickiness decline. For the AL model with a FA, a slight decline in the investment adjustment cost parameter is observed. The degree of habit formation in consumption is estimated at a somewhat higher level. The parameter of the interest rate smoothing in the policy rule increases whereas the

²⁸ Apart from the financial sector, the model is largely based on [Smets and Wouters \(2007\)](#) and [Slobodyan and Wouters \(2012b\)](#). Therefore, in a version without a FA, I can observe a very similar pattern in deviations of the parameters estimated under “AR(2)+const” learning from the parameters obtained under RE.

Table 3

Comparison of RE and alternative learning models in terms of the estimated parameters (Posterior Mode).

Parameters / Poster.Mode		RE		AR(2)+const		AR(2)		All states	
		FA	noFA	FA	noFA	FA	noFA	FA	noFA
<i>St.er shocks</i>									
Exo.risk prem.	σ_b	2.9133	2.6325	1.7918	1.3622	1.8087	1.6461	3.7498	2.4496
Investment	σ_q	0.5283	0.4761	0.3959	0.4452	0.4041	0.4525	0.9005	0.911
Price markup	σ_p	0.1624	0.1642	0.1852	0.1926	0.188	0.1895	0.1347	0.1198
Wage markup	σ_w	0.2255	0.2279	0.2277	0.2238	0.2279	0.2233	0.2564	0.2631
<i>AR. coeff-s</i>									
Exo.risk prem.	ρ_b	0.1091	0.2197	0.3881	0.4515	0.3897	0.4293	0.2318	0.368
Investment	ρ_q	0.6753	0.6887	0.4265	0.4303	0.4192	0.3924	0.4233	0.4633
Price markup	ρ_p	0.9006	0.8783	0.3206	0.1698	0.1628	0.1637	0.1518	0.1514
Wage markup	ρ_w	0.944	0.9681	0.5702	0.5357	0.553	0.7273	0.8625	0.7815
Pr.markup,ma	μ_p	0.7881	0.757	0.5020	0.3886	0.4123	0.3919	0.4447	0.4937
W.markup,ma	μ_w	0.8896	0.9234	0.4520	0.4265	0.4334	0.594	0.7772	0.6359
<i>Str. params</i>									
Adjust. cost	ϕ	6.0523	5.2727	5.9774	3.299	5.7383	3.8392	5.6242	4.4685
Habit	η	0.6765	0.7477	0.7194	0.6631	0.7201	0.6895	0.8347	0.7855
Calvo wages	ξ_w	0.7807	0.7825	0.7329	0.7498	0.7375	0.7373	0.8262	0.8452
Calvo prices	ξ_p	0.6947	0.6822	0.5756	0.5603	0.575	0.5546	0.7295	0.7041
Index. wages	ι_w	0.4472	0.424	0.2020	0.2015	0.1933	0.2014	0.3695	0.4453
Index. prices	ι_p	0.1822	0.1727	0.4184	0.4316	0.4501	0.4529	0.5504	0.5057
Int.rate smooth	ρ_r	0.8580	0.8603	0.9209	0.9216	0.9198	0.9241	0.8814	0.8814
Pol rule, inflat.	τ_π	1.9251	1.8771	1.6692	1.7382	1.6609	1.6409	1.6396	1.6545
Cap./net worth	k/N	3.4950	–	3.4525	–	3.499	–	3.50	–
Survival rate	κ	0.937	–	0.9447	–	0.9431	–	0.90	–
Elast.risk prem	el	0.013	–	0.0197	–	0.0187	–	0.0047	–
AL param.	ρ	–	–	0.9881	0.9868	0.9872	0.9917	0.8458	0.8896

parameter measuring the response to changes in inflation shows a decline. Thus under learning, the interest rate tends to adjust more gradually in response to shocks and changes in inflation. The dynamics of the interest rate under learning, combined with the higher habit formation parameter, determines more sluggish adjustments of consumption. We may conclude that learning represents an important source of endogenous inertia, but it can only partially substitute for the “mechanical” source of rigidities and the persistence of some of the disturbances.

The results presented in Tables 2 and 3 also demonstrate how the FA influences the parameters of models estimated under the RE and with learning. In particular, the autoregressive component of the exogenous risk premium shock shows some decline compared to its value estimated in models without FA. Such a result may suggest that introducing a FA captures some of the persistence in fluctuations in the external risk premium. However, the variability of the exogenous premium shock is actually increasing, which implies that the type of financial friction considered in this paper cannot fully explain the equity premium dynamics. In AL specifications, introducing a FA leads to a rise in the investment adjustment cost parameter (ϕ). Thus, the combination of the financial frictions with adaptively formed expectations has important consequences for the investment dynamics, which becomes more volatile.

Three parameters—capital-to-net-worth ratio, entrepreneurial survival rate and the elasticity of the external finance premium are jointly responsible for the FA effects in the model. The higher value of these parameters strengthens the impact of financial frictions on the real economy. Comparing the results for RE and AL models presented in Table 2, I can see that the estimated capital-to-net-worth ratio tends to increase under learning from 2.89 to 3.02 at the posterior mean (the confidence intervals overlap considerably, however). The confidence interval of the elasticity of the risk premium also rise slightly under learning. Thus, it appears that there might be a tendency for these financial parameters to trend up under learning when agents use a limited information set in forecasting equations.²⁹ Our estimates of the elasticity (0.0186 in the posterior mean for the RE model and 0.0196 for AL) are somewhat lower compared to the regression and calibration results from the previous literature.³⁰ In particular, Bernanke et al. (1999) calibrates $el=0.05$ based on a realistic value of monitoring costs and bankruptcy rates. Christensen and Dib (2008) estimate this parameter at 0.042. However, they calibrate the remaining financial parameters at a lower level. De Graeve (2008) reports a value of elasticity of 0.1047. At the same time, his

²⁹ There is a divergence in the estimated financial frictions parameters depending on the assumed form of learning. In particular, the elasticity of the external finance premium parameter in the “all states” model is lower than in the “AR(2)+const” model whereas capital-to-net-worth ratio is higher. As a result, financial-market frictions may have very different macroeconomic effects depending on the assumptions about the expectations.

³⁰ In fact, we can compare only with estimation results for models with RE.

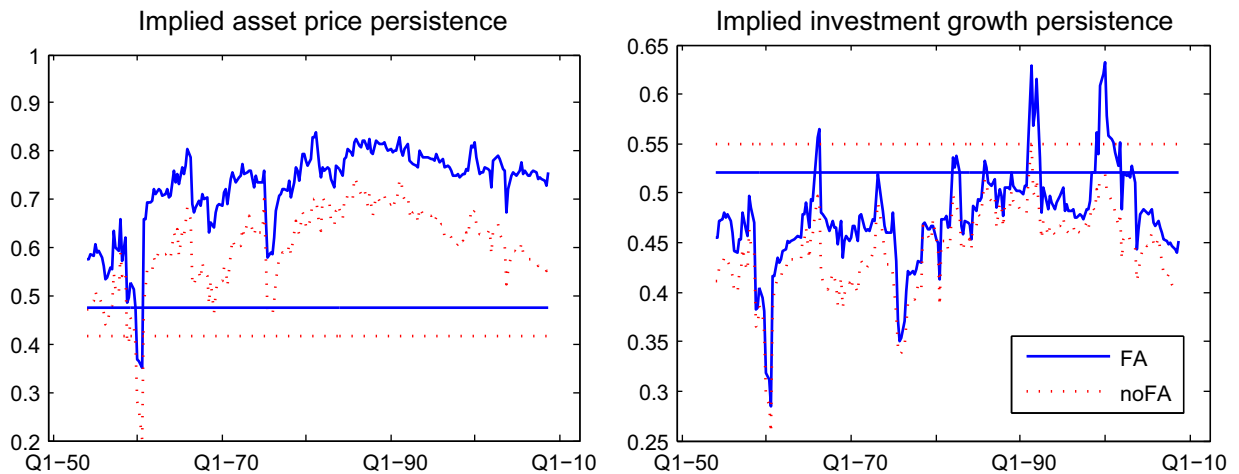


Fig. 5. Implied persistence under RE and learning.

estimated k/N ratio is twice lower compared with my results. Therefore, the estimated overall impact of financial frictions has a comparable magnitude across different studies.

An increase in k/N , the parameters elasticity as well as the adjustment cost can be explained as follows. In this model specification, the external finance premium can be decomposed into the following endogenous and exogenous components:

$$\left(E_t \hat{R}_{t+1}^K - \hat{R}_t\right) = -el \left\{ E_t \left[\hat{N}_{t+1} - \hat{Q}_t^k - \hat{k}_{t+1} \right] \right\} + \hat{b}_t. \text{ Modeling adaptive expectations in such forward-looking variables like}$$

asset prices, rental rate and inflation affects the persistence of $\hat{R}^K$ (see Eq. (10)) and $\hat{R}$ and thus the evolution of the risk premium. On another hand, the spread is largely determined by the dynamics of variables on the RHS of the equation. In particular, the net worth, asset prices and capital become more volatile under learning. Therefore, in order to match the persistence of the risk premium, a higher elasticity parameter is needed. It would be interesting to check how AL affects the parameters in the model with the other types of financial frictions and other observable (in particular, financial) variables.

4.2. Evolution of agents' beliefs and implied persistence

In this subsection, I discuss the dynamics of expectations implied by alternative learning models. I analyze the time variation of beliefs, which summarize the perceptions of agents about the economy and together with the model's structural parameters jointly determine the dynamic properties of the model. I also discuss factors affecting the estimates of the beliefs coefficients.

Fig. 4 plots the evolution of agents' beliefs given by the coefficients of the forecasting functions for the “AR(2)+constant” learning model. I present the evolution of the autoregressive component, which measures the perceived persistence and is given as a sum of AR(1) and AR(2) coefficients, and a constant. In other words, I plot agents' Perceived Law of Motion (PLM) given by (21).

Fig. 4 illustrates that beliefs about asset prices and investment (both the autoregressive parameters and the constants) show substantial time variation and appear to be most sensitive (among the rest of the forward-looking variables) to the financial market imperfections. Specifically, in the model with a FA, the perceived asset price persistence is higher than the persistence estimated in the version without financial frictions. In addition, agents believe that imperfect financial markets made the investment process more persistent in the 1950s, 1970s and again in the 1980s–1990s compared to a frictionless economy. The perceived asset price persistence evolved in the 0.4–0.8 range, with noticeable spikes in the 1980s, 1990s and 2000–s. Such dynamics appears to be consistent with developments on financial markets observed during this period. In particular, the NBER recessions were recorded around 1980–1982, 1990–1992 and 2000–2002. Prior to recessions, asset prices showed an upward trend (in particular, in the middle of the 1990s) and, generally tended to be higher than the predicted values. Positive forecasting errors motivated optimistic expectations of households and lead to the upward revision of their beliefs about asset price persistence. Such an update, in turn, contributed to the further growth of asset prices and investment. As a result, in a model with AL and FA, sudden negative shocks to asset prices would be realized in the state of high asset price and investment persistence. Therefore, such shocks may have greater real effects. During the crisis, when asset prices rapidly declined, the perceived persistence decreased as well.³¹

³¹ Slobodyan and Wouters (2012b) explain in greater detail the beliefs' updating process. In particular, for the constant term, higher (lower) than expected realizations of the forward variable result in upward (downward) revision in the constant of the forecasting equation. The updating of persistence has a state-dependent nature. For example, in periods when the inflation rate is high, revisions in persistence are positively correlated with the inflation innovations. At the same time, when a positive inflation innovation is realized in a low inflation state, the perceived inflation persistence will decline.

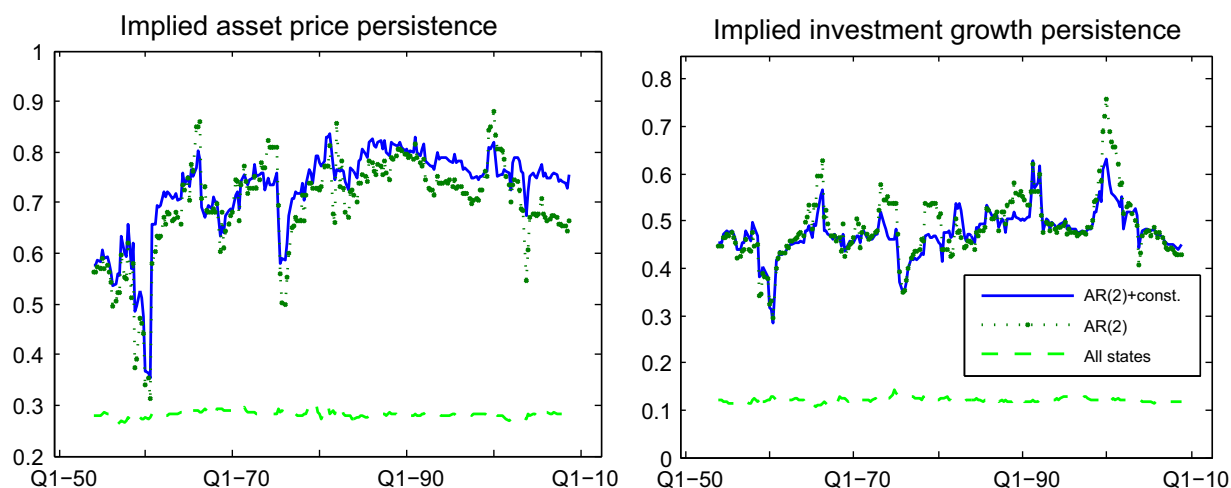


Fig. 6. Implied persistence for alternative learning models with a financial accelerator.

It is beyond the scope of this paper to assess the role of such factors as expansionary monetary policy, non-fundamental developments on the housing markets, loose credit conditions, etc. on the rapid growth in the asset prices prior to the crises of 1980s, 1990s, 2000s and quantify their contribution to the evolution of corresponding beliefs. I suppose that the shocks that transmit through the FA mechanism, i.e. the risk premium, as well as investment and monetary policy shocks, could be the main drivers behind fluctuations in the perceived asset price persistence, especially in the second half of the sample.

In addition to the autoregressive components, agents also revise their beliefs about the expected means of forward-looking variables (given by a constant in the forecasting equations). The presence of a constant brings additional changes to the transmission mechanism compared to the one implied by the RE model, where all the real variables are assumed to have a common trend growth rate and inflation is centred around a fixed inflation objective. [Slobodyan and Wouters \(2012b\)](#) interpret the variations of the constant as deviations of agents' expectations from these steady-state values. [Fig. 4](#) illustrates that constants vary the most for asset prices and investment and reflect rather a cyclical pattern of change in these variables. The fluctuations of the constant for the expected asset prices and investment rate are more pronounced in the model with a FA. Significant shifts in the expected means can add to the macroeconomic volatility and contribute to over-optimistic or -pessimistic developments in agents' expectations. Financial frictions do not have important implications for the perceived persistence of other variables—inflation, real consumption, wage and labor—making my results in this respect very similar to [Slobodyan and Wouters' findings \(2012b\)](#).³²

In order to demonstrate more explicitly the joint impact of the financial frictions and adaptive expectations on the transmission mechanism, I compute the persistence implied by the Actual Law of Motion, given by (24). The results are presented in [Fig. 5](#). The horizontal solid line shows the persistence implied by the RE model with a FA, whereas the horizontal dotted line depicts the corresponding value for the model without financial frictions. Thus, I compare the implied persistence for RE and “AR(2)+constant” learning models with and without a FA. [Fig. 5](#) demonstrates that introducing a FA resulted in higher implied asset price persistence during the whole time span. Implied investment growth persistence was also generally higher in a model with financial frictions and differentiated the most from the “noFA” level in the 1970s and after the 1990s.

[Fig. 6](#) compares the implied persistence for alternative AL schemes, which differ in terms of the variables on the RHS of the forecasting equations: “AR(2)+constant”, “AR(2)” and “all states” (near MSV). [Fig. 6](#) demonstrates that the information set used by agents to make the forecasts has important implications for the implied persistence and thus for the transmission mechanism and the business cycle. In particular, forecasts based on small learning models lead to higher implied persistence. The major difference is observed for the persistence of asset prices and investment, two variables which play a crucial role in generating FA effects. In particular, the forecasting model that incorporates all the state variables on the RHS implies a very low persistence of asset prices and thus would fail to generate significant real effects following financial shocks. The results presented in [Fig. 6](#) also indicate that introducing a constant into the forecasting function leads to a smoother transition of agents' beliefs and implied persistence. In a learning model without a constant, agents will associate any developments in observable variables with a change in their persistence, whereas in the “AR(2)+constant” model, some of the volatility may be attributed to a variation of the expected mean. As a result, the “AR(2)” model can generate greater

³² In particular, agents perceive real consumption, labor, wages and investment as highly persistent processes with relatively stable autoregressive parameters. At the same time, similar to [Slobodyan and Wouters \(2012b\)](#), we find that Kalman filter learning introduces significant time variation in agents' beliefs about inflation. Perceived inflation persistence displayed peaks around the mid 1970s and again around 1980, then gradually declined to a level around 0.6 since the mid 1980s. [Slobodyan and Wouters \(2012b\)](#) also illustrate that perceived inflation persistence is mainly driven by innovations to the price mark-up, wage mark-up and productivity shocks, which had the most pronounced impact on beliefs in the first half of the sample when inflation was relatively high.

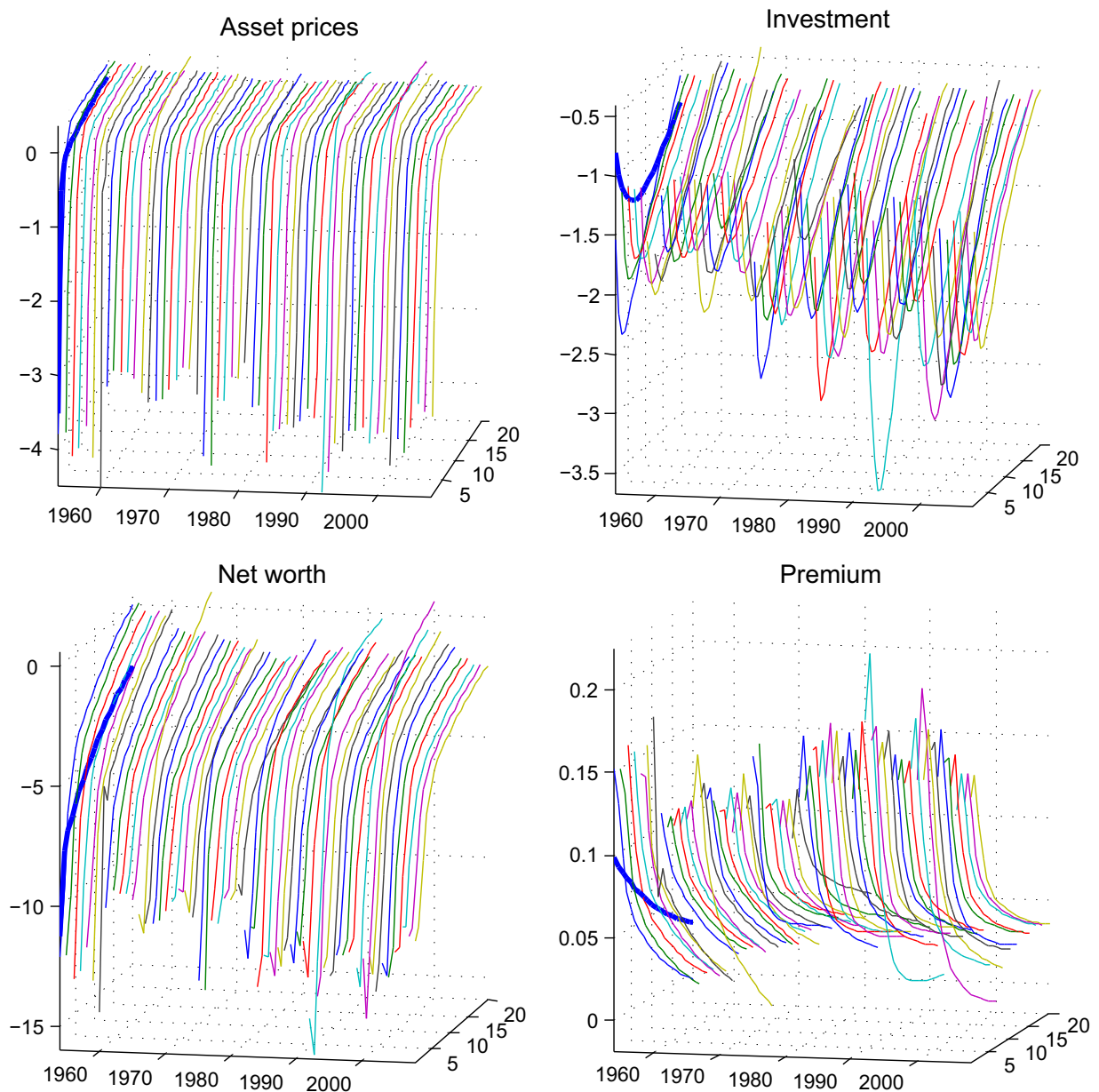


Fig. 7. Impulse responses to a risk premium shock.

swings in implied persistence and thus more volatile model dynamics. In the specification with a FA, such extra volatility can make the numerical problems related to the projection facilities more severe.

Implied persistence is an important determinant of the real effects of shocks hitting the economy. In the FA framework, agents' perceptions about financial variables such as asset prices may have important macroeconomic implications. If agents perceive asset prices to be more persistent, financial shocks will have a greater impact on households' financial position (net worth) and hence the external finance premium. Therefore, financial disturbances will involve higher cumulative effects on investment and output. The results presented in this subsection indicate that a learning model with "AR(2)+constant" beliefs may have significant implications for the shock transmission due to a higher implied persistence of asset prices as well as investment relative to the model with RE and also compared to the version without financial frictions.

To illustrate the connection between the implied persistence and the transmission mechanism, Fig. 7 compares the impulse responses to the risk premium shock for the RE and AL model based on "AR(2)+constant" forecasting functions. In fact, the figure presents the time variation of impulse responses and thus reflects the dynamic transmission mechanism under learning. The very first impulse response (denoted by the thick line) corresponds to the reaction under RE. Fig. 7 shows that the profile of asset price responses under learning depends crucially on the perceived persistence of asset prices (see Fig. 4 that shows the evolution of

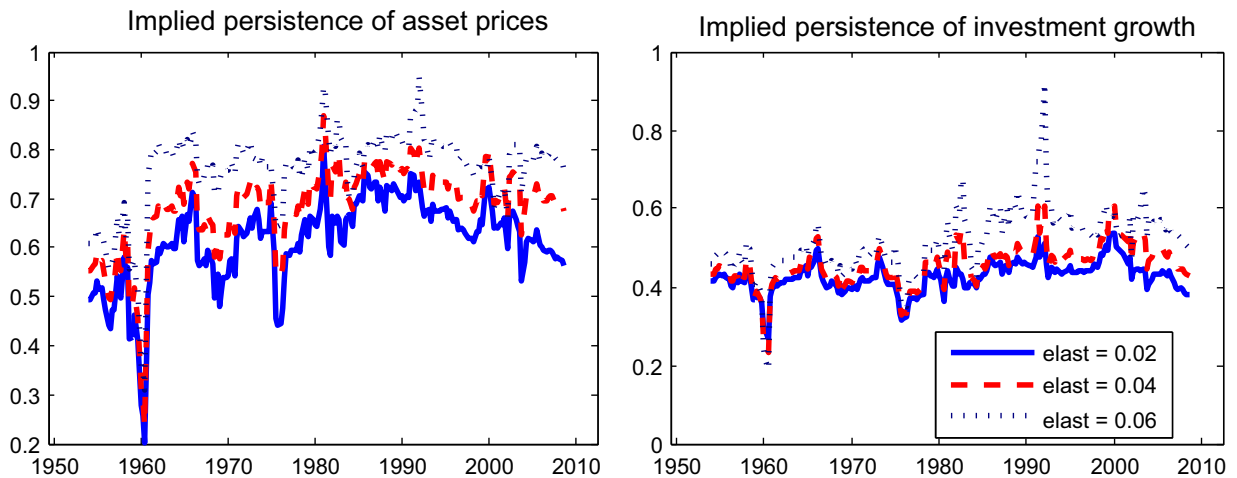


Fig. 8. Implied persistence for alternative degrees of financial frictions in the “AR(2)+const.” learning model.

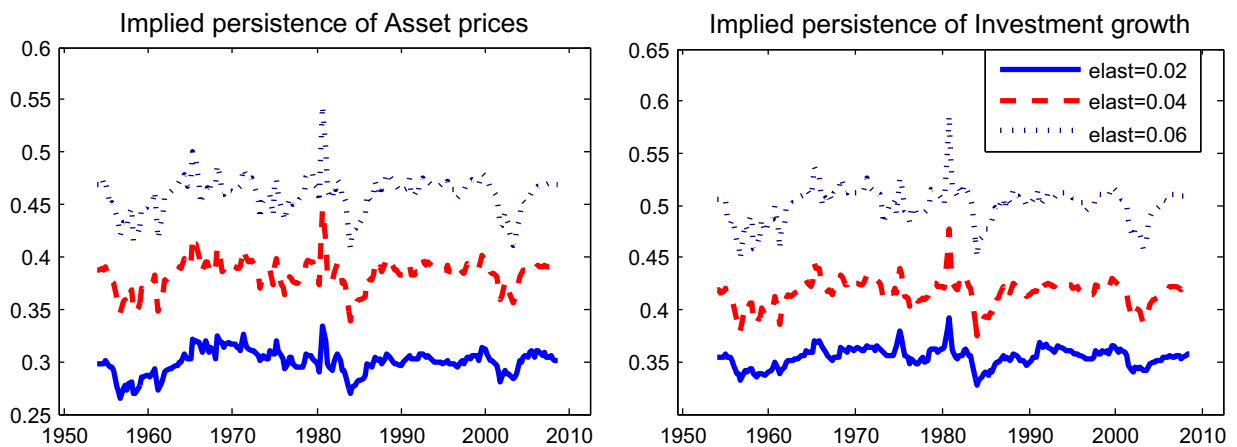


Fig. 9. Implied persistence for alternative degrees of financial frictions in the “all states” learning model.

beliefs). The peaks in the perceived asset price persistence observed in 1974, the 1980s and the 1990s lead to a dramatic fall in asset prices, which sharply reduces the net worth during these periods. As a result, even under a relatively low estimated sensitivity of the premium to changes in the entrepreneurial financial health, the gap between the cost of external financing and the risk-free rate shows a significant increase and therefore leads to a stronger impact of the FA on the real economy. The reaction of output (not shown here) in the AL model is also stronger relative to the model with RE.

The analysis of the transmission mechanism performed in this section enables several important conclusions to be drawn. First of all, learning agents perceive the developments on financial markets as being more persistent and thus more important for explaining the business cycle. Exogenous investment shocks become less important in explaining the dynamics of investments under learning. Instead, endogenous developments in asset prices driven by expectations of the private agents become essential for the evolution of investment. The profile of output responses is also closely correlated with the reactions of asset prices. However, a more persistent response on financial markets can be partly induced by structural parameters (financial friction parameters as well as adjustment costs rise slightly under learning).

In a model with RE, FA generally affects the model dynamics via amplification of the size of the responses to shocks, not necessarily affecting the level of economic persistence. In particular, introducing FA into a model with perfectly rational agents leads to a lower implied persistence of inflation and investment compared to the model without financial frictions. AL operates mainly through the (time-varying) persistence. The FA mechanism in a model with learning can generate periods with the higher persistence in asset prices, investments and output than the model without FA. FA and AL act in a mutually-reinforcing fashion, leading to a significant amplification of the real effects of shocks. Thus, a model with financial frictions and with learning may go further to explaining the output and investment dynamics when financial shocks play a dominant role in driving macroeconomic fluctuations.

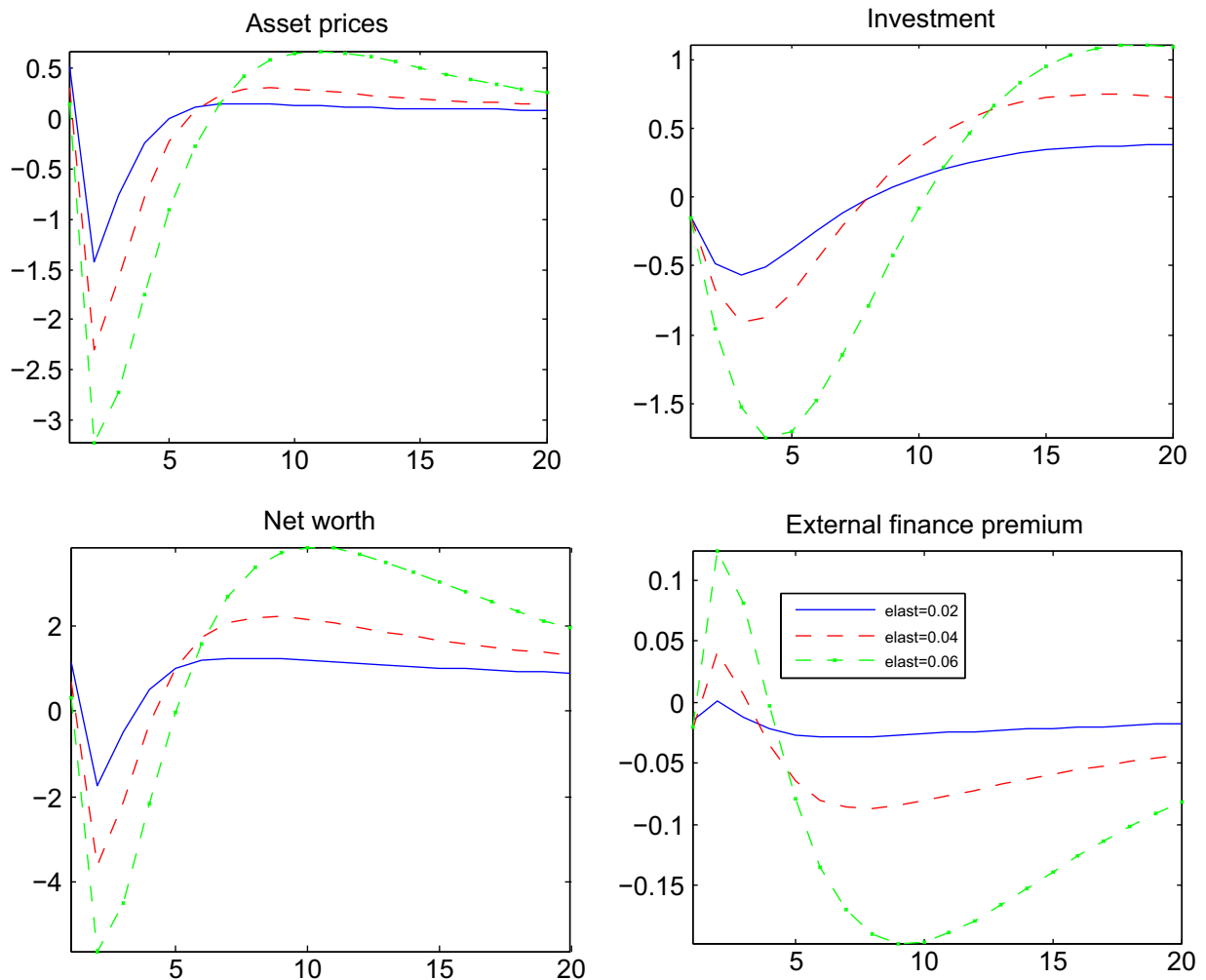


Fig. 10. Difference in responses to a risk premium shock between the “AR(2)+const.” learning and RE models for alternative degrees of financial frictions.

4.3. Simulation exercises and sensitivity analysis

In the previous subsection, I analyzed the implications of introducing financial frictions into the model with AL on the basis of the estimated values of the parameters. At the same time, the reported empirical values of the level of financial frictions vary across different studies and may depend on the estimation sample and modeling assumptions. Moreover, the inclusion of observations that cover the most recent financial distress followed by adverse macroeconomic consequences would definitely imply higher estimated values of financial parameters. Therefore, in this part of the paper, I conduct a sensitivity analysis and investigate the impact of the gradual increase in financial frictions on the macroeconomy. In order to perform this task, I fix the parameters for each model at the corresponding posterior mode value and simulate the models for 219 periods, coinciding with the duration of the estimation sample. Additionally, I assume $k/N = 2.5$ (somewhat lower than the estimated value) in order to avoid the problem with projection facilities, which may arise due to the excessive volatility of beliefs under high values of financial-friction parameters. I vary the elasticity of the risk premium: 0.02 (relatively low), 0.04 (average) and 0.06 (relatively high). In each case, I simulate agents' beliefs and calculate the implied (simulated) persistence of inflation, asset prices, growth of output and investment. In order to assess the potential ability of learning to amplify business cycle fluctuations, I compute the difference in the impulse responses to a risk premium shock between the rational expectations and AL models. I contrast the results for the two AL models that differ in terms of the information sets used by agents to forecast forward-looking variables.

Figs. 8 and 9 show the implied persistence for the “AR(2)+const.” and “all states” learning models for different degrees of financial frictions. The graphs illustrate that the implied inflation persistence is almost independent of the elasticity of the risk premium parameter. At the same time, for the AR(2)+const. learning model, higher financial frictions significantly affect the persistence of asset prices, investment and output growth. In particular, a greater elasticity of the risk premium

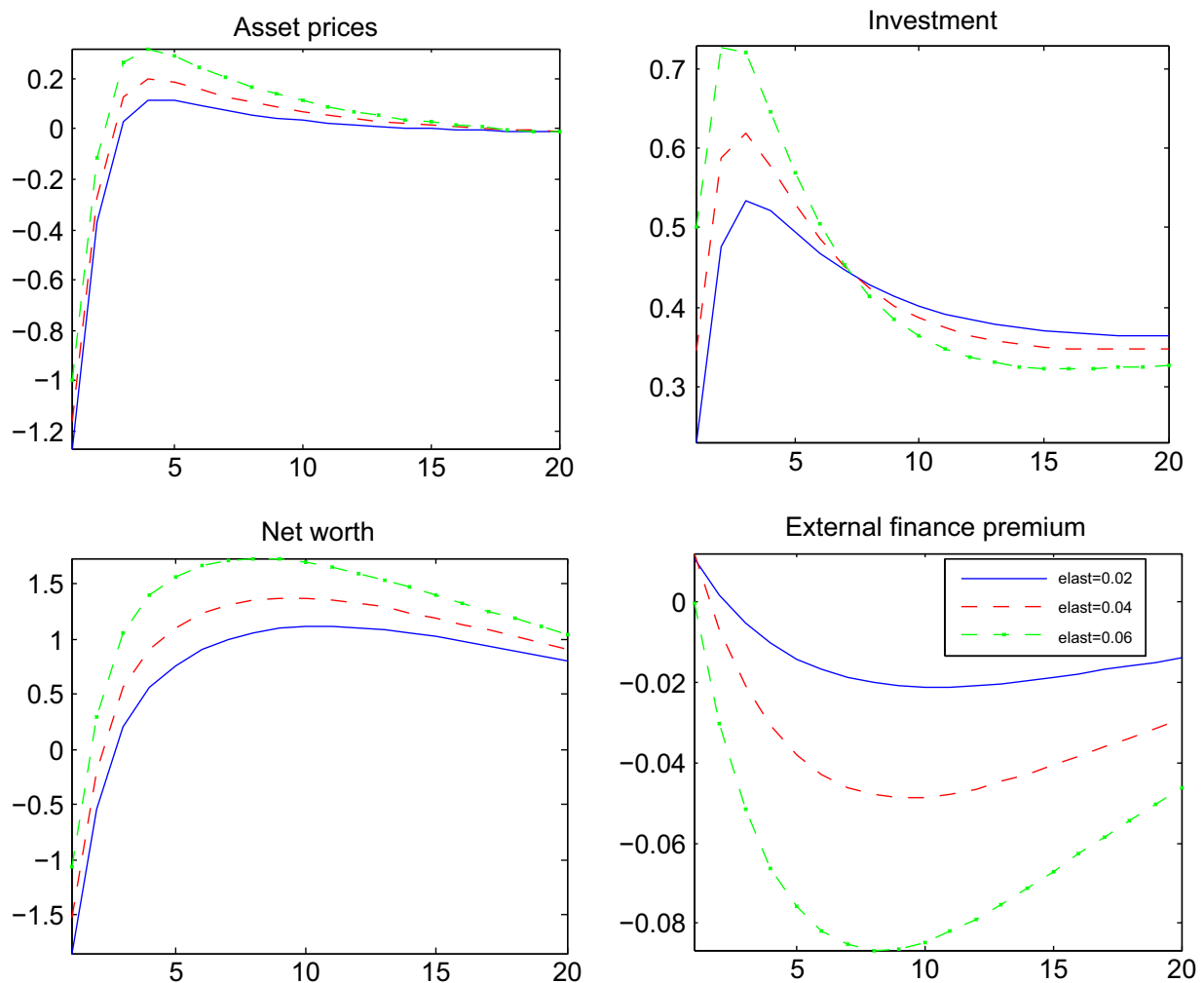


Fig. 11. Difference in responses to a risk premium shock between the "all states" learning and RE models for alternative degrees of financial frictions.

increases the level and/or time variation of the implied persistence of these variables, especially on the second half of the sample. The implied persistence simulated for the learning model in which agents employ all the state variables in forecasting functions is generally lower and more stable over time relative to the learning scheme based on small forecasting functions. Specifically, Fig. 9 demonstrates that an increase in financial frictions does lead to a somewhat higher level but hardly affects the time variation of the implied persistence in the "all states" model.

Figs. 10 and 11 present the differences in the impulse responses to a risk premium shock between the rational expectations and two alternative learning models. The greater the deviation of the line from the zero level, the stronger the reaction of the economy under learning relative to the model with RE. For example, negative values for differences in the responses of investment (Fig. 10) mean that the fall in investment was more pronounced under learning. Fig. 10 illustrates that the peak response of all the variables under learning was stronger relative to the responses under RE. The difference in the responses increases as financial frictions become stronger. For the highest value of the parameter elasticity (0.06), the increase in the external finance premium and the corresponding decline in investment is much bigger under learning compared to the model with RE. Fig. 11 demonstrates that the "all states" learning algorithm is rather unsuccessful in amplifying the business-cycle fluctuations even for the highest degree of financial frictions. In particular, under this learning scheme, the risk premium shock leads to a fall in investment that is lower (in absolute value) relative to the negative investment response under RE. Even assuming the highest value of the elasticity parameter does not reverse the results. Therefore, it appears that the type of learning model, and in particular the information set used by learning agents in forecasting, is a more fundamental factor in generating additional macroeconomic volatility than the degree of financial frictions as such.

5. Conclusions and future research

In this paper, I compare the implications of a FA mechanism for the real economy in models with alternative assumptions about expectation formation. I perform a Bayesian estimation of a medium-scale DSGE model with financial frictions assuming, on the one hand, complete rationality of expectations and, alternatively, several forms of AL that differ in terms of the information set used by agents to form their predictions. I evaluate and compare the model fit, estimated parameters and the transmission mechanism. The estimation results suggest that both financial frictions and adaptively formed expectations based on very simple forecasting functions add to improved model fit, at least on certain time intervals.

I show that the implications of a FA for the business cycle may vary depending on the expectation assumptions (RE or forms of learning). The results suggest that a learning scheme based on small forecasting functions is able to amplify the effects of financial frictions relative to a model with RE. I show that the model dynamics under learning are driven to a significant extent by the time variation of agents' beliefs about the evolution of financial variables. Specifically, I demonstrate that perceived asset price persistence in a learning model with simple forecasting equations varies through the cycle and thus differs significantly from the levels implied by the RE and alternative learning schemes. During periods when agents perceive asset prices as being relatively more persistent, shocks that affect this variable lead to more pronounced macro-economic outcomes. The asset price persistence appears to be particularly important for explaining investment dynamics. This effect is clearly observed in impulse responses. In particular, increased asset price persistence implies a more pronounced (and persistent) response of investment under the risk premium or monetary policy shocks. Therefore, I argue that certain forms of AL may play a significant role in driving and amplifying macro-economic fluctuations; it introduces an important time variation and strengthens the real effects of the FA compared to the assumption of RE. Simulation exercises illustrate that the amplification effect rises more than proportionally as financial frictions become more severe. At the same time, a learning specification in which agents use more information to generate predictions (close to MSV learning) produces very different asset price and investment dynamics. In such a framework, learning cannot significantly alter the real effects of financial frictions implied by the RE model.

This paper's findings make it possible to draw several conclusions relevant for DSGE modeling and policy analysis. In particular, due to the ability to amplify macroeconomic fluctuations, learning can be a suitable framework to simulate financial crisis scenarios and various policy reactions. Comparison of the data fit for alternative models suggests that AL with a FA represents the best specification to describe the data-generating process and analyze the shock transmission in the second half of the sample. In addition, the results imply that the link between asset prices and the real economy has become more important and the sensitivity of the economy to financial shocks increased after the middle 1980s. Such an empirical conclusion is supported by the impulse responses of real variables that show higher time variation in the 1990s and 2000s following risk premium shock. Which economic processes or policy reactions could contribute to the increased propagation of financial shocks is an important question for further research. A stable economic environment with low interest rates and inflation could be partly responsible for the adverse dynamics of asset prices and the development of a bubble leading up to the crisis. Future research could complement this paper with an analysis of monetary policy in the economy with AL and financial frictions. One could study how a strong anti-inflationary stance or expansionary monetary policy can impact the implied persistence of asset prices as well as the variation of real variables and inflation. In addition, it is possible to conduct an experiment on policy rules that incorporate the response to asset prices and examine whether such rules could deliver better macroeconomic outcomes in terms of monetary and financial stability.

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Appendix A

A.1. Comparing the forecasting performance

Table 4 compares the 1-step ahead root mean squared forecast error for different forecast samples under RE and the best-performing AL model. In out-of-sample exercise, I re-estimate the models every four quarters. The biggest gain of the learning model relative to the RE model is obtained for inflation and investment growth. I also compute the RMSE statistics based on in-sample predictions in order to evaluate the forecasts implied by learning model estimated over the full sample.

A.2. Robustness check

In this section, I report the results of the robustness analysis. In particular, I check the sensitivity of the results to the alternative specifications of the PLM and initialization of the learning mechanism.

Table 4
Forecast Comparison of the RE and AL Models.

	π	Cons.	Inv.	GDP	Wage	Hours
Out-of-sample: 1990q1–2008q3						
1q RMSE, RE	0.25	0.52	1.41	0.55	0.61	0.53
% gain(+)/loss(–) of AL	4.51	2.56	5.21	3.84	–0.87	3.74
In sample: 1970q1–2008q3						
1q RMSE, RE	0.29	0.57	1.74	0.71	0.54	0.53
% gain(+)/loss(–) of AL	4.83	–2.67	1.44	–1.35	–2.47	–0.90
In sample: 1990q1–2008q3						
1q RMSE, RE	0.24	0.48	1.39	0.53	0.61	0.45
% gain(+)/loss(–) of AL	5.76	–1.73	2.87	1.29	–1.15	1.14
In sample: 1990q1–2011q3						
1q RMSE, RE	0.24	0.63	1.90	0.67	0.62	0.61
% gain(+)/loss(–) of AL	–1.34	3.62	4.92	3.11	–0.77	3.75

Table 5
Robustness Analysis.

Model specification	Marg.Lik. (Laplace approx.)
AR(2)+constant (baseline)	–1199.72
AR(1)+const. +lag inflation	–1198.44
AR(1)+const. +lag inflation, output, interest rate	–1201.40
AR(2)+const. +(marg.cost,invest.,interest rate)	–1205.16
Baseline + alternative REE-consistent initial beliefs	–1200.57
Baseline + individual ρ	–1205.47
All states + shocks	–1220.41

First of all, I examine the performance of AL under alternative assumptions about the form of the forecasting equations. I consider the following small-to-medium-scale PLM: an AR(1) process plus lagged inflation; an AR(1) process plus lagged inflation, interest rate and output³³; AR(2) process with additional RHS variables which could be relevant for explaining the dynamics of forward variables (I experiment with the combinations of the following variables: marginal cost for inflation, investment for price of capital, interest rate for consumption). In addition, I consider a specification with the broader information set, which includes all the state variables and also shocks.

Secondly, I evaluate the performance of AL under the alternative formulation of the initial beliefs. More specifically, I assume that learning algorithm starts from less informative initial beliefs, which are consistent with REE obtained with coefficients set at their prior means. Finally, in the baseline AL specification, the parameter ρ , which measures the persistence of beliefs, was restricted to be identical for all the forward-looking variables. Here, I relax this assumption and estimate the model with seven ρ parameters. This allows agents to adapt the speed of beliefs' updating in different forecasting functions.

The main results of the robustness exercise are reported in Table 5. The Table demonstrates that alternative learning specifications based on small forecasting functions produce the marginal likelihood and the model dynamics that is very similar to the baseline AR(2) model. Adding more variables on the RHS of the forecasting equations generally deteriorates the fit. In this respect, my results are consistent with Slobodyan and Wouters (2012b), who show that including too many variables into the forecasting functions leads to overfitting problem and worsens the model fit. I also find that results are robust to alternative initializations of initial beliefs consistent with REE. My results also indicate that differentiating between the AL parameters ρ leads to a higher uncertainty around the estimates of ρ and as a result to a worse marginal likelihood compared to the baseline case. The fit is however still somewhat better than under RE. Depending on the assumptions on the individual priors/calibrations of ρ the marginal likelihood ranges from –1201 to –1205.

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³³ The similar fit is obtained for the corresponding AR(2) specification, therefore not reported in the table with results.

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